BITS_Project_Fourth_Week_Tasks_Sajal (1)

April 4, 2026

1 Retail Stocking Priority Classification: Logistic Regression vs Custom Deep Learning vs Hugging Face

1.1 Project reframing

This notebook reframes the original retail demand forecasting problem into a **multi-class product stocking priority classification** problem.

Instead of asking only:

- *How many units will sell?*

we also ask:

- *Which products or categories deserve higher stocking priority?*
- *Which items are likely to generate better profit relative to shelf space?*
- *Which items should be deprioritized because of short shelf life, low margin, or supply risk?*

This reframing is closer to how retail managers think in practice.

1.2 Modeling goals

We will build and compare three model families:

1. **Logistic Regression** for multi-class classification
2. A **custom deep learning architecture**
3. A **Hugging Face pre-trained transformer** fine-tuned for classification

We will also compare:

- **text-only classification**
- **numeric-only classification**
- **hybrid text + numeric classification**

1.3 Business target

We create a derived target called **Stocking Priority Class** with three classes:

- `Low`
- `Medium`
- `High`

This class is based on a business-oriented score that combines:

- profit density
- sales demand
- shelf life
- fill rate
- supplier delay
- backorder risk

1.4 Important note

The target used in this notebook is **derived from business rules**, not directly provided in the dataset. That means the models are learning to approximate a business prioritization rule, not a historical label already present in the data.

1.5 Mathematical formulation

We define the following intermediate quantities:

1.5.1 1. Unit Margin

$$\text{Unit Margin} = \text{Current Price} - \text{Unit Cost}$$

1.5.2 2. Gross Profit

$$\text{Gross Profit} = \text{Unit Margin} \times \text{Daily Units Sold}$$

1.5.3 3. Profit Density

$$\text{Profit Density} = \frac{\text{Gross Profit}}{\text{Shelf Capacity}}$$

1.5.4 4. Stocking Priority Score

After min-max normalization of relevant variables, we define:

$$\text{Priority Score} = 0.40 \cdot \text{Norm}(\text{Profit Density}) + 0.20 \cdot \text{Norm}(\text{Daily Units Sold}) + 0.15 \cdot \text{Norm}(\text{Fill Rate}) + 0.10 \cdot \text{Norm}(\text{She}$$

Finally, the score is split into three quantile-based classes:

- Low
- Medium
- High

1.6 Colab setup

If you are running this notebook in Google Colab and any of the NLP libraries are missing, uncomment and run the next cell once, then restart the runtime.

[1]:
```
# Optional Colab setup cell
# Uncomment only if your environment does not already have these packages.
```

```
# !pip install -q transformers datasets accelerate sentencepiece evaluate
```

[2]:
```
# ============================================================
# Imports
# ============================================================

import os
import random
import warnings
import inspect

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    f1_score,
    classification_report,
    confusion_matrix
)

warnings.filterwarnings("ignore")
plt.rcParams["figure.figsize"] = (10, 5)

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
from scipy.special import softmax
```

1.7 Load the dataset

[3]:
```
# ============================================================
# Load dataset
# ============================================================

df = pd.read_csv("/content/master_retail_dataset_v2.csv")

print("Dataset shape:", df.shape)
```

```
display(df.head())
```

```
Dataset shape: (2000, 58)

         Date  DayOfWeek  Month  Year  Is_Weekend  Is_Holiday  Product_ID  \
0  04/13/2023          3      4  2023           0           0        1002
1  12/15/2023          4     12  2023           0           0        1014
2  09/28/2023          3      9  2023           0           0        1015
3  04/17/2023          0      4  2023           0           0        1012
4  03/13/2023          0      3  2023           0           0        1016

      Category  Base_Price  Discount_Percentage  …  Online_Sales_Units  \
0      Grocery      410.09                 0.01  …                  12
1         Home       89.98                 0.30  …                  31
2  Electronics      404.88                 0.14  …                  32
3      Grocery      347.97                 0.43  …                  32
4      Grocery      280.16                 0.14  …                  23

   In_Store_Sales_Units App_Traffic_Index  Website_Visits  Backorder_Flag  \
0                    71               488             896               N
1                    71               619            1406               N
2                    57               626            1432               N
3                    66               626            1432               N
4                    77               564            1191               N

   Supplier_Delay_Days  Fill_Rate_Pct  Loyalty_Program_Usage_Count  \
0                    0          88.93                           11
1                    2          90.16                           31
2                    6          88.00                           23
3                    2          91.17                            7
4                    0          89.39                            9

   Repeat_Purchase_Rate  Avg_Basket_Size
0                  0.14             3.77
1                  0.14             1.48
2                  0.38             1.07
3                  0.13             4.67
4                  0.16             2.17

[5 rows x 58 columns]
```

1.8 Inspect columns and data types

[4]:
```
# Basic inspection
print("\nColumns:")
print(df.columns.tolist())

print("\nData types:")
```

```
display(df.dtypes)
```

```
Columns:
['Date', 'DayOfWeek', 'Month', 'Year', 'Is_Weekend', 'Is_Holiday', 'Product_ID',
'Category', 'Base_Price', 'Discount_Percentage', 'Current_Price', 'Store_ID',
'Store_Type', 'Competitor_Price', 'Footfall_Index', 'Avg_Temperature',
'Rainfall_mm', 'Social_Media_Sentiment', 'Lead_Time_Days', 'Safety_Stock_Level',
'Stock_On_Hand', 'Daily_Units_Sold', 'Festival_Name', 'Festival_Type',
'UPC_EAN', 'Product_Name', 'Sub_Category', 'Brand_Name', 'Brand_Tier',
'Shelf_Life_Days', 'No_of_Checkout_Counters', 'Avg_Billing_Time_min',
'Local_Population_Density', 'Product_Age_Days', 'No_of_Customer_Purchases',
'Unit_Cost', 'Promotional_Campaign_Flag', 'Competitor_Promotion_Flag',
'Google_Trends_Current_Wk', 'Google_Trends_Lag_1w', 'Shelf_Capacity',
'Promotion_Type', 'Marketing_Spend', 'Payday_Flag', 'School_Vacation_Flag',
'Local_Event_Flag', 'Launch_Date', 'Seasonal_Product_Flag',
'Online_Sales_Units', 'In_Store_Sales_Units', 'App_Traffic_Index',
'Website_Visits', 'Backorder_Flag', 'Supplier_Delay_Days', 'Fill_Rate_Pct',
'Loyalty_Program_Usage_Count', 'Repeat_Purchase_Rate', 'Avg_Basket_Size']

Data types:

Date                            object
DayOfWeek                        int64
Month                            int64
Year                             int64
Is_Weekend                       int64
Is_Holiday                       int64
Product_ID                       int64
Category                        object
Base_Price                     float64
Discount_Percentage            float64
Current_Price                  float64
Store_ID                         int64
Store_Type                      object
Competitor_Price               float64
Footfall_Index                 float64
Avg_Temperature                float64
Rainfall_mm                    float64
Social_Media_Sentiment         float64
Lead_Time_Days                   int64
Safety_Stock_Level               int64
Stock_On_Hand                    int64
Daily_Units_Sold                 int64
Festival_Name                   object
Festival_Type                   object
UPC_EAN                          int64
Product_Name                    object
```

```
Sub_Category                    object
Brand_Name                      object
Brand_Tier                      object
Shelf_Life_Days                  int64
No_of_Checkout_Counters          int64
Avg_Billing_Time_min           float64
Local_Population_Density         int64
Product_Age_Days                 int64
No_of_Customer_Purchases         int64
Unit_Cost                      float64
Promotional_Campaign_Flag       object
Competitor_Promotion_Flag       object
Google_Trends_Current_Wk         int64
Google_Trends_Lag_1w             int64
Shelf_Capacity                   int64
Promotion_Type                  object
Marketing_Spend                  int64
Payday_Flag                      int64
School_Vacation_Flag             int64
Local_Event_Flag                 int64
Launch_Date                     object
Seasonal_Product_Flag           object
Online_Sales_Units               int64
In_Store_Sales_Units             int64
App_Traffic_Index                int64
Website_Visits                   int64
Backorder_Flag                  object
Supplier_Delay_Days              int64
Fill_Rate_Pct                  float64
Loyalty_Program_Usage_Count      int64
Repeat_Purchase_Rate           float64
Avg_Basket_Size                float64
dtype: object
```

1.9 Clean selected flag columns

Some business flags are stored as text. We convert them into numeric 0/1 form where needed.

[5]:
```
# ============================================================
# Clean selected binary-like flags
# ============================================================

binary_map = {
    "yes": 1, "y": 1, "true": 1, "1": 1,
    "no": 0, "n": 0, "false": 0, "0": 0
}
```

```
for col in ["Backorder_Flag", "Promotional_Campaign_Flag",␣
 ↪"Competitor_Promotion_Flag", "Seasonal_Product_Flag"]:
    if col in df.columns:
        df[col] = (
            df[col]
            .astype(str)
            .str.strip()
            .str.lower()
            .map(binary_map)
            .fillna(0)
            .astype(int)
        )

# Convert launch date to datetime
df["Launch_Date"] = pd.to_datetime(df["Launch_Date"], errors="coerce")
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")

display(df[["Backorder_Flag", "Promotional_Campaign_Flag",␣
 ↪"Competitor_Promotion_Flag", "Seasonal_Product_Flag"]].head())
```

```
   Backorder_Flag  Promotional_Campaign_Flag  Competitor_Promotion_Flag  \
0               0                          0                          0
1               0                          1                          1
2               0                          0                          1
3               0                          1                          0
4               0                          0                          1

   Seasonal_Product_Flag
0                      1
1                      1
2                      0
3                      0
4                      0
```

1.10 Create the business target: Stocking Priority Class

[6]:
```
# ============================================================
# Create business-oriented target
# ============================================================

# 1. Unit margin
df["Unit_Margin"] = df["Current_Price"] - df["Unit_Cost"]

# 2. Gross profit
df["Gross_Profit"] = df["Unit_Margin"] * df["Daily_Units_Sold"]

# 3. Profit density relative to shelf capacity
```

```python
df["Profit_Density"] = df["Gross_Profit"] / df["Shelf_Capacity"].replace(0, np.
↪nan)
df["Profit_Density"] = df["Profit_Density"].fillna(0)

# Helper function for min-max normalization
def minmax_scale(series):
    series = series.astype(float)
    mn = series.min()
    mx = series.max()
    if mx == mn:
        return pd.Series(np.zeros(len(series)), index=series.index)
    return (series - mn) / (mx - mn)

# Normalized components
df["Norm_Profit_Density"] = minmax_scale(df["Profit_Density"])
df["Norm_Daily_Units_Sold"] = minmax_scale(df["Daily_Units_Sold"])
df["Norm_Fill_Rate"] = minmax_scale(df["Fill_Rate_Pct"])
df["Norm_Shelf_Life"] = minmax_scale(df["Shelf_Life_Days"])
df["Norm_Supplier_Delay"] = minmax_scale(df["Supplier_Delay_Days"])

# Business-oriented priority score
df["Stocking_Priority_Score"] = (
    0.40 * df["Norm_Profit_Density"]
    + 0.20 * df["Norm_Daily_Units_Sold"]
    + 0.15 * df["Norm_Fill_Rate"]
    + 0.10 * df["Norm_Shelf_Life"]
    + 0.10 * (1 - df["Norm_Supplier_Delay"])
    + 0.05 * (1 - df["Backorder_Flag"])
)

# Convert score into 3 balanced classes
df["Stocking_Priority_Class"] = pd.qcut(
    df["Stocking_Priority_Score"],
    q=3,
    labels=["Low", "Medium", "High"]
)

print("Class distribution:")
display(df["Stocking_Priority_Class"].value_counts())
```

```
Class distribution:

Stocking_Priority_Class
Low       667
High      667
Medium    666
Name: count, dtype: int64
```

1.11 Initial business analysis

The next cells generate actual analysis outputs for:

1. Which categories appear most often in the **High** stocking priority class
2. Which categories have the shortest average shelf life
3. Which products rank highest on the stocking-priority score

[7]:
```
# ============================================================
# Category-level analysis
# ============================================================

category_summary = (
    df.groupby("Category")
      .agg(
          Records=("Category", "size"),
          Avg_Daily_Units_Sold=("Daily_Units_Sold", "mean"),
          Avg_Current_Price=("Current_Price", "mean"),
          Avg_Unit_Margin=("Unit_Margin", "mean"),
          Avg_Gross_Profit=("Gross_Profit", "mean"),
          Avg_Shelf_Capacity=("Shelf_Capacity", "mean"),
          Avg_Shelf_Life_Days=("Shelf_Life_Days", "mean")
      )
)

category_high_share = (
    df.assign(Is_High=(df["Stocking_Priority_Class"] == "High").astype(int))
      .groupby("Category")["Is_High"]
      .mean()
      .rename("High_Priority_Share")
)

category_summary = category_summary.join(category_high_share)
category_summary = category_summary.sort_values("High_Priority_Share",␣
 ↪ascending=False)

display(category_summary.round(3))
```

```
             Records  Avg_Daily_Units_Sold  Avg_Current_Price  \
Category
Home             391               111.591            187.388
Clothing         378               113.257            184.610
Electronics      387               110.687            194.809
Beauty           402               112.244            182.500
Grocery          442               113.405            180.041

             Avg_Unit_Margin  Avg_Gross_Profit  Avg_Shelf_Capacity  \
Category
Home                  16.937          1372.081              70.987
```

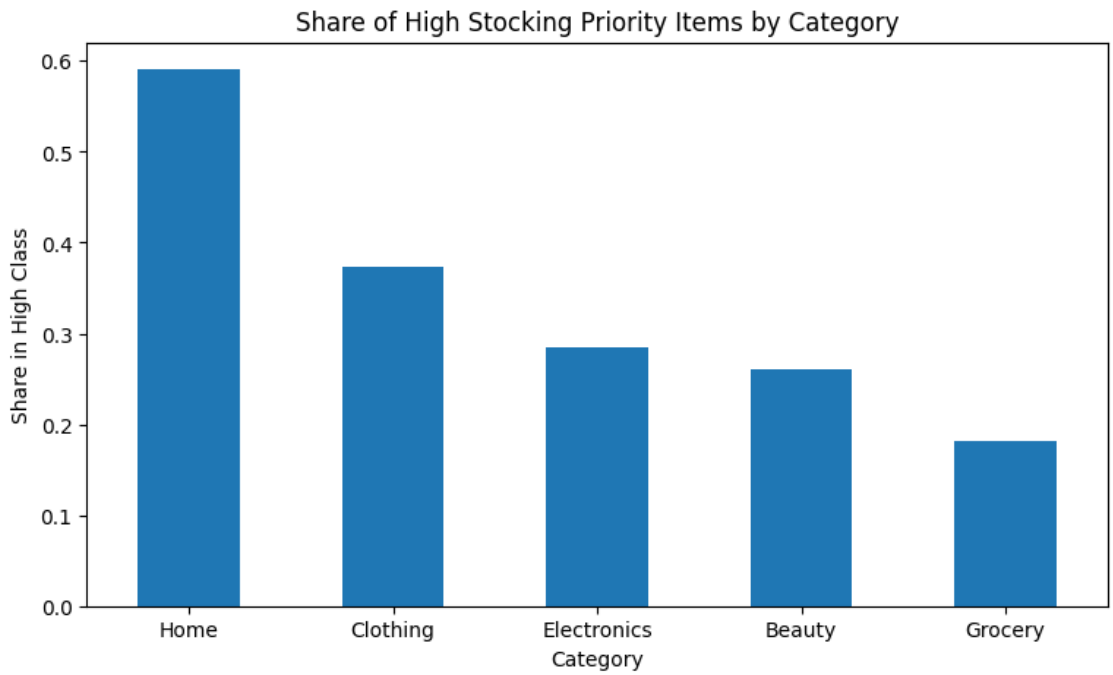
```
Clothing              16.219          1340.999              69.765
Electronics           18.332          1412.922              73.829
Beauty                20.036          1690.405              68.888
Grocery               17.507          1577.671              71.068

             Avg_Shelf_Life_Days  High_Priority_Share
Category
Home                    2222.056                0.591
Clothing                 943.532                0.373
Electronics             1394.491                0.284
Beauty                   530.913                0.261
Grocery                   42.649                0.181
```

1.11.1 Interpretation

The table above identifies which categories are more likely to appear in the **High** stocking priority class under the business rule defined in this notebook.

In this dataset, categories such as **Home**, **Clothing**, **Electronics**, and **Beauty** tend to perform better than **Grocery** on the stocking-priority score. Grocery remains important for sales, but its much shorter shelf life can reduce its stocking attractiveness under this business framing.

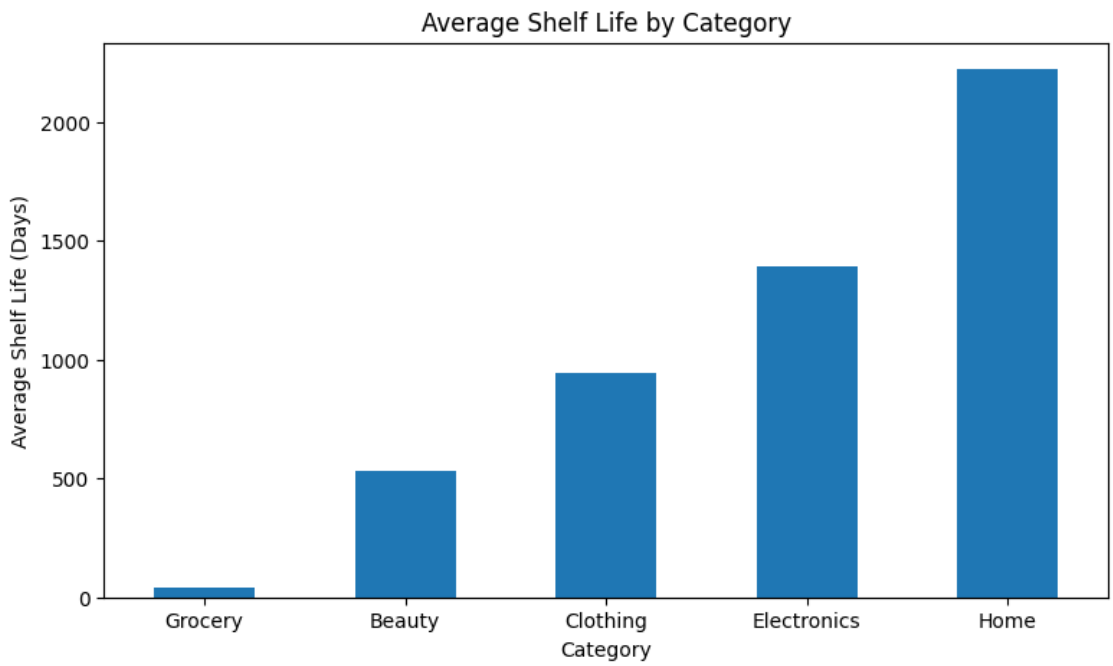
[8]:
```
# Plot the share of High priority items by category
high_share_plot = category_summary["High_Priority_Share"].
 ↪sort_values(ascending=False)

plt.figure(figsize=(9, 5))
high_share_plot.plot(kind="bar")
plt.title("Share of High Stocking Priority Items by Category")
plt.ylabel("Share in High Class")
plt.xlabel("Category")
plt.xticks(rotation=0)
plt.show()
```



[9]:
```
# ============================================================
# Shelf-life analysis
# ============================================================

shelf_life_by_category = (
    df.groupby("Category")["Shelf_Life_Days"]
      .mean()
      .sort_values()
)

display(shelf_life_by_category.round(2))

plt.figure(figsize=(9, 5))
shelf_life_by_category.plot(kind="bar")
plt.title("Average Shelf Life by Category")
plt.ylabel("Average Shelf Life (Days)")
plt.xlabel("Category")
plt.xticks(rotation=0)
plt.show()
```

```
Category
Grocery          42.65
Beauty          530.91
Clothing        943.53
Electronics    1394.49
```

```
Home            2222.06
Name: Shelf_Life_Days, dtype: float64
```



1.11.2 Interpretation

Shelf life is central to the instructor's framing.

If a category has short shelf life, heavy stocking can create spoilage risk and reduce business attractiveness even when demand is strong. In this dataset, **Grocery** has by far the shortest average shelf life, which makes it more operationally risky than categories such as **Home** or **Electronics**.

[10]:
```
# ============================================================
# Top products by average stocking priority score
# ============================================================

top_products = (
    df.groupby(["Category", "Product_Name"])
      .agg(
          Avg_Priority_Score=("Stocking_Priority_Score", "mean"),
          Avg_Profit_Density=("Profit_Density", "mean"),
          Avg_Shelf_Life=("Shelf_Life_Days", "mean"),
          Records=("Product_Name", "size")
      )
      .sort_values("Avg_Priority_Score", ascending=False)
      .head(15)
)
```

```
display(top_products.round(3))
```

```
                                    Avg_Priority_Score  Avg_Profit_Density  \
Category Product_Name
Home     Philips Kitchenware 1004                0.561              27.981
         Tupperware Bedding 1001                 0.558              53.273
         Tupperware Bedding 1017                 0.555              -3.550
         Sleepwell Furniture 1013                0.550               8.842
         Sleepwell Kitchenware 1015              0.549              22.536
         Ikea Furniture 1018                     0.540              59.867
         Tupperware Decor 1014                   0.538              32.426
         Tupperware Decor 1005                   0.537              25.452
         Tupperware Bedding 1006                 0.536              18.324
         Ikea Bedding 1008                       0.536              51.706
         Ikea Kitchenware 1010                   0.533              33.997
         Tupperware Decor 1003                   0.532              16.206
         Ikea Kitchenware 1007                   0.528             -14.767
         Sleepwell Furniture 1000                0.524              27.497
Clothing FabIndia Formalwear 1006                0.524              63.025

                                    Avg_Shelf_Life  Records
Category Product_Name
Home     Philips Kitchenware 1004           2366.0       20
         Tupperware Bedding 1001            2790.0       21
         Tupperware Bedding 1017            2810.0       19
         Sleepwell Furniture 1013           2850.0       19
         Sleepwell Kitchenware 1015         2782.0       19
         Ikea Furniture 1018                2770.0       16
         Tupperware Decor 1014              2244.0       20
         Tupperware Decor 1005              2280.0       18
         Tupperware Bedding 1006            2358.0       15
         Ikea Bedding 1008                  2326.0       11
         Ikea Kitchenware 1010              2038.0       25
         Tupperware Decor 1003              2178.0       19
         Ikea Kitchenware 1007              2470.0       24
         Sleepwell Furniture 1000           1117.0       11
Clothing FabIndia Formalwear 1006           1057.0       15
```

1.12 Build a column-aware text template

Instead of using a raw combined text field, we now build a more meaningful text template that preserves the role of each textual attribute.

1.12.1 Example template

```
Category: Beauty | Product: Lakme Makeup 1009 | Brand: Lakme | Brand Tier:
Premium | Promotion: Flash Sale | Festival: Diwali
```

This is more informative than simply concatenating tokens without context.

[11]:
```
# ============================================================
# Create column-aware text template
# ============================================================

text_columns_for_template = [
    "Category", "Product_Name", "Sub_Category", "Brand_Name",
    "Brand_Tier", "Store_Type", "Promotion_Type", "Festival_Name",
    "Festival_Type"
]

for col in text_columns_for_template:
    df[col] = df[col].fillna("None").astype(str)

df["text_template"] = (
    "Category: " + df["Category"]
    + " | Product: " + df["Product_Name"]
    + " | SubCategory: " + df["Sub_Category"]
    + " | Brand: " + df["Brand_Name"]
    + " | BrandTier: " + df["Brand_Tier"]
    + " | StoreType: " + df["Store_Type"]
    + " | Promotion: " + df["Promotion_Type"]
    + " | FestivalName: " + df["Festival_Name"]
    + " | FestivalType: " + df["Festival_Type"]
)

display(df[["text_template"]].head(3))
```

```
                                        text_template
0  Category: Grocery | Product: Nestle Beverages …
1  Category: Home | Product: Tupperware Decor 101…
2  Category: Electronics | Product: Sony Computer…
```

1.13 Define label encoder and train/validation/test splits

We use a **stratified** split so that the class balance is preserved.

This is important because the task is now **multi-class classification**.

[12]:
```
# ============================================================
# Encode target labels
# ============================================================

label_encoder = LabelEncoder()
df["label_id"] = label_encoder.fit_transform(df["Stocking_Priority_Class"])

# Build Python-native label dictionaries for compatibility with
# Hugging Face model configuration (JSON serializable keys/values).
```

```
class_names = [str(x) for x in label_encoder.classes_]
label_mapping = {str(label): int(i) for i, label in enumerate(class_names)}
inverse_label_mapping = {int(i): str(label) for i, label in␣
 ↪enumerate(class_names)}

print("Label mapping:", label_mapping)
print("Inverse label mapping:", inverse_label_mapping)
```

```
Label mapping: {'High': 0, 'Low': 1, 'Medium': 2}
Inverse label mapping: {0: 'High', 1: 'Low', 2: 'Medium'}
```

[13]:

```
# ============================================================
# Train / validation / test split
# ============================================================

train_df, temp_df = train_test_split(
    df,
    test_size=0.30,
    random_state=SEED,
    stratify=df["label_id"]
)

val_df, test_df = train_test_split(
    temp_df,
    test_size=0.50,
    random_state=SEED,
    stratify=temp_df["label_id"]
)

print("Train shape:", train_df.shape)
print("Validation shape:", val_df.shape)
print("Test shape:", test_df.shape)

print("\nTrain class distribution:")
display(train_df["Stocking_Priority_Class"].value_counts(normalize=True).
 ↪round(3))

print("\nValidation class distribution:")
display(val_df["Stocking_Priority_Class"].value_counts(normalize=True).round(3))

print("\nTest class distribution:")
display(test_df["Stocking_Priority_Class"].value_counts(normalize=True).
 ↪round(3))
```

```
Train shape: (1400, 70)
Validation shape: (300, 70)
Test shape: (300, 70)
```

```
Train class distribution:

Stocking_Priority_Class
Low       0.334
High      0.334
Medium    0.333
Name: proportion, dtype: float64


Validation class distribution:

Stocking_Priority_Class
Low       0.333
Medium    0.333
High      0.333
Name: proportion, dtype: float64


Test class distribution:

Stocking_Priority_Class
Low       0.333
Medium    0.333
High      0.333
Name: proportion, dtype: float64
```

1.14 Prepare numeric features

To avoid trivial leakage, we **exclude** the exact variables used to construct the target score whenever possible.

We keep context-rich numeric variables such as: - traffic - seasonality proxies - marketing - store operations - trends - customer behavior

[14]:
```
# ============================================================
# Numeric feature selection
# ============================================================

numeric_feature_cols = [
    "DayOfWeek",
    "Month",
    "Year",
    "Is_Weekend",
    "Is_Holiday",
    "Base_Price",
    "Discount_Percentage",
    "Competitor_Price",
    "Footfall_Index",
    "Avg_Temperature",
    "Rainfall_mm",
    "Lead_Time_Days",
```

```
    "Safety_Stock_Level",
    "Stock_On_Hand",
    "No_of_Checkout_Counters",
    "Avg_Billing_Time_min",
    "Local_Population_Density",
    "Product_Age_Days",
    "No_of_Customer_Purchases",
    "Promotional_Campaign_Flag",
    "Competitor_Promotion_Flag",
    "Google_Trends_Current_Wk",
    "Google_Trends_Lag_1w",
    "Marketing_Spend",
    "Payday_Flag",
    "School_Vacation_Flag",
    "Local_Event_Flag",
    "Seasonal_Product_Flag",
    "App_Traffic_Index",
    "Website_Visits",
    "Loyalty_Program_Usage_Count",
    "Repeat_Purchase_Rate",
    "Avg_Basket_Size"
]

# Ensure the selected columns exist
numeric_feature_cols = [col for col in numeric_feature_cols if col in df.
 ↪columns]

print("Numeric features used:", len(numeric_feature_cols))
print(numeric_feature_cols)
```

```
Numeric features used: 33
['DayOfWeek', 'Month', 'Year', 'Is_Weekend', 'Is_Holiday', 'Base_Price',
'Discount_Percentage', 'Competitor_Price', 'Footfall_Index', 'Avg_Temperature',
'Rainfall_mm', 'Lead_Time_Days', 'Safety_Stock_Level', 'Stock_On_Hand',
'No_of_Checkout_Counters', 'Avg_Billing_Time_min', 'Local_Population_Density',
'Product_Age_Days', 'No_of_Customer_Purchases', 'Promotional_Campaign_Flag',
'Competitor_Promotion_Flag', 'Google_Trends_Current_Wk', 'Google_Trends_Lag_1w',
'Marketing_Spend', 'Payday_Flag', 'School_Vacation_Flag', 'Local_Event_Flag',
'Seasonal_Product_Flag', 'App_Traffic_Index', 'Website_Visits',
'Loyalty_Program_Usage_Count', 'Repeat_Purchase_Rate', 'Avg_Basket_Size']
```

2 1. Logistic Regression Baselines

We compare three variants:

1. **Text-only logistic regression**
2. **Numeric-only logistic regression**
3. **Hybrid logistic regression (text + numeric)**

This helps us understand whether the business signal is coming more from category/product text, structured numeric context, or both together.

[15]:
```
# ============================================================
# Helper functions for evaluation
# ============================================================

def evaluate_predictions(y_true, y_pred, model_name):
    """
    Return a dictionary with common multi-class classification metrics.
    """
    return {
        "Model": model_name,
        "Accuracy": accuracy_score(y_true, y_pred),
        "Macro_F1": f1_score(y_true, y_pred, average="macro"),
        "Weighted_F1": f1_score(y_true, y_pred, average="weighted")
    }

def print_report(y_true, y_pred, title):
    print(f"\n{title}")
    print("=" * len(title))
    print(classification_report(y_true, y_pred, target_names=label_encoder.
 ↪classes_))
```

[16]:
```
# ============================================================
# Text-only logistic regression
# ============================================================

text_only_clf = Pipeline([
    ("tfidf", TfidfVectorizer(max_features=5000, ngram_range=(1, 2))),
    ("clf", LogisticRegression(max_iter=3000, class_weight="balanced",␣
 ↪random_state=SEED))
])

text_only_clf.fit(train_df["text_template"], train_df["label_id"])
text_only_pred = text_only_clf.predict(test_df["text_template"])

text_only_metrics = evaluate_predictions(test_df["label_id"], text_only_pred,␣
 ↪"LogReg_TextOnly")
print_report(test_df["label_id"], text_only_pred, "Text-only Logistic␣
 ↪Regression Report")
text_only_metrics
```

```
Text-only Logistic Regression Report
====================================
              precision    recall  f1-score   support
```

```
        High       0.52      0.49      0.50       100
         Low       0.42      0.56      0.48       100
      Medium       0.36      0.26      0.30       100

    accuracy                           0.44       300
   macro avg       0.43      0.44      0.43       300
weighted avg       0.43      0.44      0.43       300
```

[16]:
```
{'Model': 'LogReg_TextOnly',
 'Accuracy': 0.43666666666666665,
 'Macro_F1': 0.4285254597461405,
 'Weighted_F1': 0.4285254597461404}
```

[17]:
```python
# ============================================================
# Numeric-only logistic regression
# ============================================================

numeric_only_preprocessor = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

numeric_only_X_train = numeric_only_preprocessor.
 ↪fit_transform(train_df[numeric_feature_cols])
numeric_only_X_test = numeric_only_preprocessor.
 ↪transform(test_df[numeric_feature_cols])

numeric_only_clf = LogisticRegression(
    max_iter=3000,
    class_weight="balanced",
    random_state=SEED
)

numeric_only_clf.fit(numeric_only_X_train, train_df["label_id"])
numeric_only_pred = numeric_only_clf.predict(numeric_only_X_test)

numeric_only_metrics = evaluate_predictions(test_df["label_id"],␣
 ↪numeric_only_pred, "LogReg_NumericOnly")
print_report(test_df["label_id"], numeric_only_pred, "Numeric-only Logistic␣
 ↪Regression Report")
numeric_only_metrics
```

```
Numeric-only Logistic Regression Report
========================================
              precision    recall  f1-score   support
```

```
        High       0.44      0.51      0.47       100
         Low       0.45      0.56      0.50       100
      Medium       0.35      0.21      0.26       100

    accuracy                           0.43       300
   macro avg       0.41      0.43      0.41       300
weighted avg       0.41      0.43      0.41       300
```

[17]:
```
{'Model': 'LogReg_NumericOnly',
 'Accuracy': 0.4266666666666667,
 'Macro_F1': 0.4115740740740741,
 'Weighted_F1': 0.4115740740740741}
```

[18]:
```
# ============================================================
# Hybrid logistic regression (text + numeric)
# ============================================================

hybrid_preprocessor = ColumnTransformer([
    ("text", TfidfVectorizer(max_features=5000, ngram_range=(1, 2)),␣
 ↪"text_template"),
    ("num", Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler())
    ]), numeric_feature_cols)
])

hybrid_clf = Pipeline([
    ("preprocessor", hybrid_preprocessor),
    ("clf", LogisticRegression(max_iter=3000, class_weight="balanced",␣
 ↪random_state=SEED))
])

hybrid_clf.fit(train_df[["text_template"] + numeric_feature_cols],␣
 ↪train_df["label_id"])
hybrid_pred = hybrid_clf.predict(test_df[["text_template"] +␣
 ↪numeric_feature_cols])

hybrid_metrics = evaluate_predictions(test_df["label_id"], hybrid_pred,␣
 ↪"LogReg_Hybrid")
print_report(test_df["label_id"], hybrid_pred, "Hybrid Logistic Regression␣
 ↪Report")
hybrid_metrics
```

```
Hybrid Logistic Regression Report
==================================
              precision    recall  f1-score   support
```

```
        High       0.50      0.55      0.53       100
         Low       0.54      0.58      0.56       100
      Medium       0.35      0.29      0.32       100

    accuracy                           0.47       300
   macro avg       0.46      0.47      0.47       300
weighted avg       0.46      0.47      0.47       300
```

[18]:
```
{'Model': 'LogReg_Hybrid',
 'Accuracy': 0.4733333333333333,
 'Macro_F1': 0.4669826626254581,
 'Weighted_F1': 0.4669826626254581}
```

[19]:
```python
# Logistic regression comparison table
logreg_results = pd.DataFrame([
    text_only_metrics,
    numeric_only_metrics,
    hybrid_metrics
]).sort_values(["Weighted_F1", "Accuracy"], ascending=False)

display(logreg_results.round(4))
```

```
                Model  Accuracy  Macro_F1  Weighted_F1
2       LogReg_Hybrid    0.4733    0.4670       0.4670
0     LogReg_TextOnly    0.4367    0.4285       0.4285
1  LogReg_NumericOnly    0.4267    0.4116       0.4116
```

2.0.1 Interpretation

The logistic regression comparison tells us whether: - product/category text alone is enough - numeric context alone is enough - or the **hybrid view** is strongest

In many retail problems, the hybrid setup performs best because it combines: - semantic category/product information - structured operational context

3 2. Custom Deep Learning Architecture

We build a custom **hybrid neural network** with:

- a **text branch** that learns from the column-aware text template
- a **numeric branch** that learns from structured numeric features

The two branches are merged before the final softmax classification layer.

[20]:
```python
# ============================================================
# TensorFlow / Keras imports
# ============================================================
```

```python
import tensorflow as tf
from tensorflow.keras.layers import (
    Input,
    TextVectorization,
    Embedding,
    Bidirectional,
    LSTM,
    Dense,
    Dropout,
    Concatenate
)
from tensorflow.keras.models import Model
from tensorflow.keras.callbacks import EarlyStopping

tf.random.set_seed(SEED)
```

[21]:
```python
# ============================================================
# Prepare data for the custom hybrid model
# ============================================================

# Text arrays
X_train_text = train_df["text_template"].astype(str).values
X_val_text = val_df["text_template"].astype(str).values
X_test_text = test_df["text_template"].astype(str).values

# Numeric arrays
numeric_imputer = SimpleImputer(strategy="median")
numeric_scaler = StandardScaler()

X_train_num = numeric_imputer.fit_transform(train_df[numeric_feature_cols])
X_val_num = numeric_imputer.transform(val_df[numeric_feature_cols])
X_test_num = numeric_imputer.transform(test_df[numeric_feature_cols])

X_train_num = numeric_scaler.fit_transform(X_train_num)
X_val_num = numeric_scaler.transform(X_val_num)
X_test_num = numeric_scaler.transform(X_test_num)

# Labels
y_train = train_df["label_id"].values
y_val = val_df["label_id"].values
y_test = test_df["label_id"].values

print("Text training shape:", X_train_text.shape)
print("Numeric training shape:", X_train_num.shape)
```

```
Text training shape: (1400,)
Numeric training shape: (1400, 33)
```

[22]:
```
# ============================================================
# Text vectorization layer
# ============================================================

max_tokens = 8000
sequence_length = 40

text_vectorizer = TextVectorization(
    max_tokens=max_tokens,
    output_mode="int",
    output_sequence_length=sequence_length
)

# Adapt vectorizer only on training text
text_vectorizer.adapt(X_train_text)
```

[23]:
```
# ============================================================
# Build custom hybrid neural network
# ============================================================

# Text input branch
text_input = Input(shape=(1,), dtype=tf.string, name="text_input")
x_text = text_vectorizer(text_input)
x_text = Embedding(input_dim=max_tokens, output_dim=64)(x_text)
x_text = Bidirectional(LSTM(32))(x_text)
x_text = Dense(32, activation="relu")(x_text)
x_text = Dropout(0.3)(x_text)

# Numeric input branch
num_input = Input(shape=(X_train_num.shape[1],), name="numeric_input")
x_num = Dense(64, activation="relu")(num_input)
x_num = Dropout(0.3)(x_num)
x_num = Dense(32, activation="relu")(x_num)

# Merge both branches
x = Concatenate()([x_text, x_num])
x = Dense(64, activation="relu")(x)
x = Dropout(0.3)(x)
output = Dense(len(label_encoder.classes_), activation="softmax")(x)

custom_model = Model(inputs=[text_input, num_input], outputs=output)

custom_model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
custom_model.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
text_input (InputLayer)	(None, 1)	0	-
text_vectorization (TextVectorization)	(None, 40)	0	text_input[0][0]
embedding (Embedding)	(None, 40, 64)	512,000	text_vectorizati…
numeric_input (InputLayer)	(None, 33)	0	-
bidirectional (Bidirectional)	(None, 64)	24,832	embedding[0][0]
dense_1 (Dense)	(None, 64)	2,176	numeric_input[0]…
dense (Dense)	(None, 32)	2,080	bidirectional[0]…
dropout_1 (Dropout)	(None, 64)	0	dense_1[0][0]
dropout (Dropout)	(None, 32)	0	dense[0][0]
dense_2 (Dense)	(None, 32)	2,080	dropout_1[0][0]
concatenate (Concatenate)	(None, 64)	0	dropout[0][0], dense_2[0][0]
dense_3 (Dense)	(None, 64)	4,160	concatenate[0][0]
dropout_2 (Dropout)	(None, 64)	0	dense_3[0][0]
dense_4 (Dense)	(None, 3)	195	dropout_2[0][0]

Total params: 547,523 (2.09 MB)

Trainable params: 547,523 (2.09 MB)

```
Non-trainable params: 0 (0.00 B)
```

[24]:
```python
# ============================================================
# Train custom hybrid neural network
# ============================================================

early_stop = EarlyStopping(
    monitor="val_loss",
    patience=3,
    restore_best_weights=True
)

history = custom_model.fit(
    {"text_input": X_train_text, "numeric_input": X_train_num},
    y_train,
    validation_data=(
        {"text_input": X_val_text, "numeric_input": X_val_num},
        y_val
    ),
    epochs=8,
    batch_size=32,
    callbacks=[early_stop],
    verbose=1
)
```

```
Epoch 1/8
44/44              15s 89ms/step -
accuracy: 0.3557 - loss: 1.1380 - val_accuracy: 0.4200 - val_loss: 1.0824
Epoch 2/8
44/44              3s 66ms/step -
accuracy: 0.4279 - loss: 1.0575 - val_accuracy: 0.4333 - val_loss: 1.0395
Epoch 3/8
44/44              4s 90ms/step -
accuracy: 0.4443 - loss: 1.0251 - val_accuracy: 0.4567 - val_loss: 1.0256
Epoch 4/8
44/44              4s 87ms/step -
accuracy: 0.4914 - loss: 0.9849 - val_accuracy: 0.4200 - val_loss: 1.0244
Epoch 5/8
44/44              4s 80ms/step -
accuracy: 0.4971 - loss: 0.9723 - val_accuracy: 0.4067 - val_loss: 1.0272
Epoch 6/8
44/44              5s 114ms/step -
accuracy: 0.5079 - loss: 0.9705 - val_accuracy: 0.4267 - val_loss: 1.0235
Epoch 7/8
44/44              4s 84ms/step -
accuracy: 0.5336 - loss: 0.9492 - val_accuracy: 0.4300 - val_loss: 1.0261
```

```
Epoch 8/8
44/44              2s 52ms/step -
accuracy: 0.5293 - loss: 0.9379 - val_accuracy: 0.3900 - val_loss: 1.0287
```

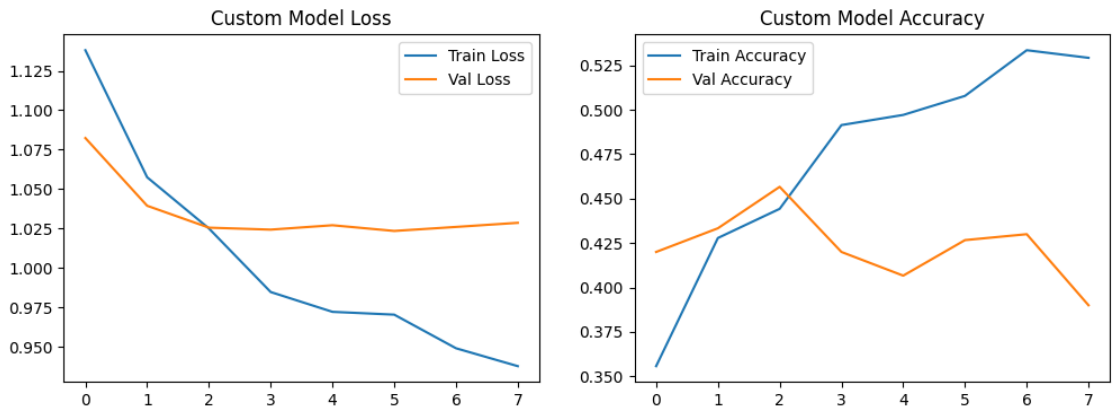
[25]:
```
# ============================================================
# Evaluate custom hybrid neural network
# ============================================================

custom_pred_probs = custom_model.predict(
    {"text_input": X_test_text, "numeric_input": X_test_num},
    verbose=0
)

custom_pred = np.argmax(custom_pred_probs, axis=1)

custom_metrics = evaluate_predictions(y_test, custom_pred, "CustomDL_Hybrid")
print_report(y_test, custom_pred, "Custom Deep Learning Hybrid Model Report")
custom_metrics
```

```
Custom Deep Learning Hybrid Model Report
========================================
              precision    recall  f1-score   support

        High       0.52      0.62      0.57       100
         Low       0.47      0.63      0.54       100
      Medium       0.43      0.20      0.27       100

    accuracy                           0.48       300
   macro avg       0.47      0.48      0.46       300
weighted avg       0.47      0.48      0.46       300
```

[25]:
```
{'Model': 'CustomDL_Hybrid',
 'Accuracy': 0.48333333333333334,
 'Macro_F1': 0.45892680922035134,
 'Weighted_F1': 0.4589268092203513}
```

[26]:
```
# Plot training curves
history_df = pd.DataFrame(history.history)

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].plot(history_df["loss"], label="Train Loss")
axes[0].plot(history_df["val_loss"], label="Val Loss")
axes[0].set_title("Custom Model Loss")
axes[0].legend()
```

```
axes[1].plot(history_df["accuracy"], label="Train Accuracy")
axes[1].plot(history_df["val_accuracy"], label="Val Accuracy")
axes[1].set_title("Custom Model Accuracy")
axes[1].legend()

plt.show()
```



4 3. Hugging Face Pre-trained Transformer

For the Hugging Face baseline, we fine-tune a pre-trained text classification model on the **column-aware text template**.

To keep training practical in Colab, we use a compact transformer such as:

- `distilbert-base-uncased`

This model is **text-only**, so it does not use the numeric feature block.

[27]:
```
# ============================================================
# Hugging Face imports
# ============================================================

from datasets import Dataset
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
    DataCollatorWithPadding,
    TrainingArguments,
    Trainer
)
```

[28]:
```
# ============================================================
# Prepare Hugging Face datasets
```

```
# ============================================================

hf_train_df = train_df[["text_template", "label_id"]].
 ↪rename(columns={"text_template": "text", "label_id": "label"})
hf_val_df = val_df[["text_template", "label_id"]].
 ↪rename(columns={"text_template": "text", "label_id": "label"})
hf_test_df = test_df[["text_template", "label_id"]].
 ↪rename(columns={"text_template": "text", "label_id": "label"})

hf_train = Dataset.from_pandas(hf_train_df.reset_index(drop=True))
hf_val = Dataset.from_pandas(hf_val_df.reset_index(drop=True))
hf_test = Dataset.from_pandas(hf_test_df.reset_index(drop=True))
```

[29]:
```
# ============================================================
# Tokenizer and tokenization
# ============================================================

checkpoint = "distilbert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(checkpoint)

def tokenize_batch(batch):
    return tokenizer(batch["text"], truncation=True)

tokenized_train = hf_train.map(tokenize_batch, batched=True)
tokenized_val = hf_val.map(tokenize_batch, batched=True)
tokenized_test = hf_test.map(tokenize_batch, batched=True)

data_collator = DataCollatorWithPadding(tokenizer=tokenizer)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
faster downloads.

config.json:   0%|          | 0.00/483 [00:00<?, ?B/s]

tokenizer_config.json:   0%|          | 0.00/48.0 [00:00<?, ?B/s]

vocab.txt: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

Map:   0%|          | 0/1400 [00:00<?, ? examples/s]

Map:   0%|          | 0/300 [00:00<?, ? examples/s]

Map:   0%|          | 0/300 [00:00<?, ? examples/s]
```

[30]:
```
# ============================================================
# Load pre-trained model
```

```
# ==============================================================

hf_model = AutoModelForSequenceClassification.from_pretrained(
    checkpoint,
    num_labels=len(class_names),
    id2label=inverse_label_mapping,
    label2id=label_mapping
)
```

```
model.safetensors:   0%|          | 0.00/268M [00:00<?, ?B/s]

Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_transform.weight   | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |
pre_classifier.bias      | MISSING    |
classifier.weight        | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the
checkpoint. Consider training on your downstream task.
```

[31]:
```
# ==============================================================
# Metrics for Hugging Face Trainer
# ==============================================================

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=1)
    return {
        "accuracy": accuracy_score(labels, preds),
        "macro_f1": f1_score(labels, preds, average="macro"),
        "weighted_f1": f1_score(labels, preds, average="weighted")
    }
```

[32]:
```
# ==============================================================
# Training arguments
# ==============================================================
```

```python
# Transformers API differs across versions. This block detects which
# arguments are supported and builds a compatible TrainingArguments object.
training_kwargs = {
    "output_dir": "hf_stocking_priority_results",
    "learning_rate": 2e-5,
    "per_device_train_batch_size": 16,
    "per_device_eval_batch_size": 16,
    "num_train_epochs": 2,
    "weight_decay": 0.01,
    "report_to": "none",
    "seed": SEED
}

supported_args = set(inspect.signature(TrainingArguments.__init__).parameters.
 ↪keys())

if "evaluation_strategy" in supported_args:
    training_kwargs["evaluation_strategy"] = "epoch"
if "save_strategy" in supported_args:
    training_kwargs["save_strategy"] = "no"
elif "save_steps" in supported_args:
    training_kwargs["save_steps"] = 10_000

if "logging_strategy" in supported_args:
    training_kwargs["logging_strategy"] = "epoch"
elif "logging_steps" in supported_args:
    training_kwargs["logging_steps"] = 100

# Older Transformers versions use do_train / do_eval flags.
if "do_train" in supported_args:
    training_kwargs["do_train"] = True
if "do_eval" in supported_args:
    training_kwargs["do_eval"] = True

training_args = TrainingArguments(**training_kwargs)
training_args
```

[32]:
```
TrainingArguments(
accelerator_config={'split_batches': False, 'dispatch_batches': None,
'even_batches': True, 'use_seedable_sampler': True, 'non_blocking': False,
'gradient_accumulation_kwargs': None, 'use_configured_state': False},
adam_beta1=0.9,
adam_beta2=0.999,
adam_epsilon=1e-08,
auto_find_batch_size=False,
average_tokens_across_devices=True,
batch_eval_metrics=False,
```

```
bf16=False,
bf16_full_eval=False,
data_seed=None,
dataloader_drop_last=False,
dataloader_num_workers=0,
dataloader_persistent_workers=False,
dataloader_pin_memory=True,
dataloader_prefetch_factor=None,
ddp_backend=None,
ddp_broadcast_buffers=None,
ddp_bucket_cap_mb=None,
ddp_find_unused_parameters=None,
ddp_timeout=1800,
debug=[],
deepspeed=None,
disable_tqdm=False,
do_eval=True,
do_predict=False,
do_train=True,
enable_jit_checkpoint=False,
eval_accumulation_steps=None,
eval_delay=0,
eval_do_concat_batches=True,
eval_on_start=False,
eval_steps=None,
eval_strategy=IntervalStrategy.NO,
eval_use_gather_object=False,
fp16=False,
fp16_full_eval=False,
fsdp=[],
fsdp_config={'min_num_params': 0, 'xla': False, 'xla_fsdp_v2': False,
'xla_fsdp_grad_ckpt': False},
full_determinism=False,
gradient_accumulation_steps=1,
gradient_checkpointing=False,
gradient_checkpointing_kwargs=None,
greater_is_better=None,
group_by_length=False,
hub_always_push=False,
hub_model_id=None,
hub_private_repo=None,
hub_revision=None,
hub_strategy=HubStrategy.EVERY_SAVE,
hub_token=<HUB_TOKEN>,
ignore_data_skip=False,
include_for_metrics=[],
include_num_input_tokens_seen=no,
```

```
label_names=None,
label_smoothing_factor=0.0,
learning_rate=2e-05,
length_column_name=length,
liger_kernel_config=None,
load_best_model_at_end=False,
local_rank=-1,
log_level=passive,
log_level_replica=warning,
log_on_each_node=True,
logging_dir=None,
logging_first_step=False,
logging_nan_inf_filter=True,
logging_steps=500,
logging_strategy=IntervalStrategy.EPOCH,
lr_scheduler_kwargs=None,
lr_scheduler_type=SchedulerType.LINEAR,
max_grad_norm=1.0,
max_steps=-1,
metric_for_best_model=None,
neftune_noise_alpha=None,
num_train_epochs=2,
optim=OptimizerNames.ADAMW_TORCH_FUSED,
optim_args=None,
optim_target_modules=None,
output_dir=hf_stocking_priority_results,
parallelism_config=None,
per_device_eval_batch_size=16,
per_device_train_batch_size=16,
prediction_loss_only=False,
project=huggingface,
push_to_hub=False,
remove_unused_columns=True,
report_to=[],
restore_callback_states_from_checkpoint=False,
resume_from_checkpoint=None,
run_name=None,
save_on_each_node=False,
save_only_model=False,
save_steps=500,
save_strategy=SaveStrategy.NO,
save_total_limit=None,
seed=42,
skip_memory_metrics=True,
tf32=None,
torch_compile=False,
torch_compile_backend=None,
```

```
torch_compile_mode=None,
torch_empty_cache_steps=None,
trackio_space_id=trackio,
use_cache=False,
use_cpu=False,
use_liger_kernel=False,
warmup_ratio=None,
warmup_steps=0,
weight_decay=0.01,
)
```

[33]:
```
# ============================================================
# Trainer
# ============================================================

# Note:
# - tokenized_train / tokenized_val are already tokenized
# - data_collator handles padding
# - some Transformers versions do not accept tokenizer= in Trainer

trainer = Trainer(
    model=hf_model,
    args=training_args,
    train_dataset=tokenized_train,
    eval_dataset=tokenized_val,
    data_collator=data_collator,
    compute_metrics=compute_metrics
)
```

[34]:
```
# ============================================================
# Train Hugging Face model
# ============================================================

trainer.train()

# Explicit evaluation is more version-robust than relying only on
# evaluation_strategy.
trainer.evaluate()
```

```
<IPython.core.display.HTML object>
```

```
<IPython.core.display.HTML object>
```

[34]:
```
{'eval_loss': 1.0661290884017944,
 'eval_accuracy': 0.41,
 'eval_macro_f1': 0.37900896961792047,
 'eval_weighted_f1': 0.3790089696179204,
 'eval_runtime': 23.0003,
```

```
 'eval_samples_per_second': 13.043,
 'eval_steps_per_second': 0.826,
 'epoch': 2.0}
```

[35]:
```
# ============================================================
# Evaluate Hugging Face model on test data
# ============================================================

hf_test_pred = trainer.predict(tokenized_test)
hf_preds = np.argmax(hf_test_pred.predictions, axis=1)

hf_metrics = evaluate_predictions(hf_test_df["label"].values, hf_preds,␣
 ↪"HF_DistilBERT_TextOnly")
print_report(hf_test_df["label"].values, hf_preds, "Hugging Face DistilBERT␣
 ↪Report")
hf_metrics
```

```
<IPython.core.display.HTML object>


Hugging Face DistilBERT Report
==============================
              precision    recall  f1-score   support

        High       0.54      0.57      0.55       100
         Low       0.40      0.65      0.50       100
      Medium       0.30      0.10      0.15       100

    accuracy                           0.44       300
   macro avg       0.41      0.44      0.40       300
weighted avg       0.41      0.44      0.40       300
```

[35]:
```
{'Model': 'HF_DistilBERT_TextOnly',
 'Accuracy': 0.44,
 'Macro_F1': 0.40061942976326353,
 'Weighted_F1': 0.4006194297632635}
```

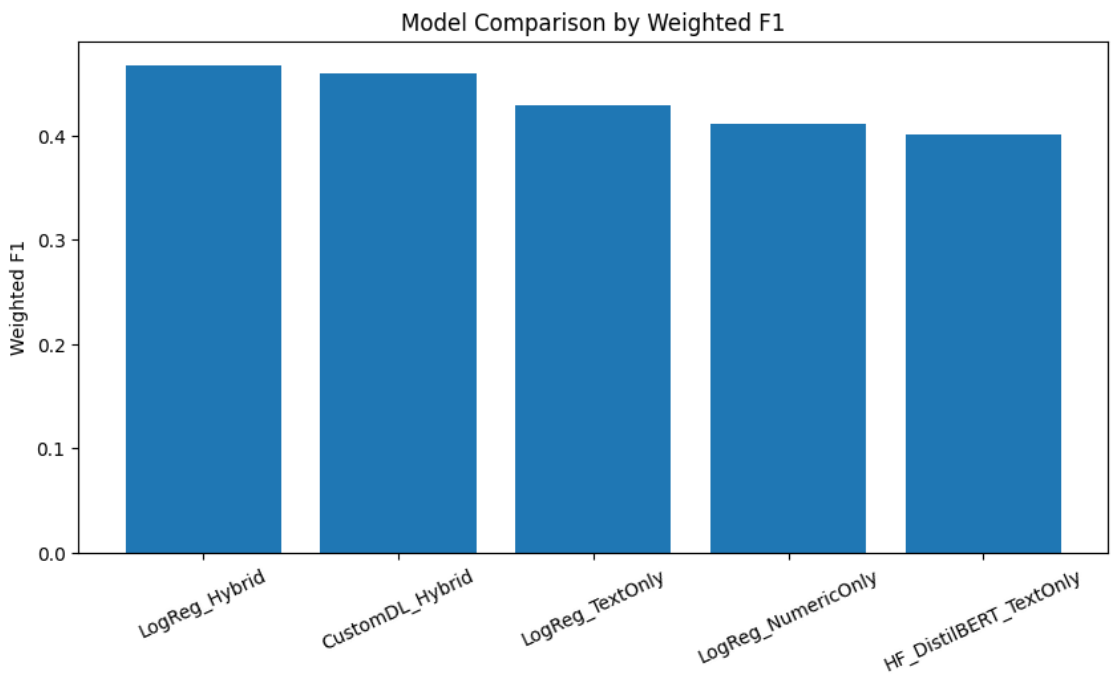
5 Final comparison: Logistic Regression vs Custom DL vs Hugging Face

[36]:
```
# ============================================================
# Combine all model results
# ============================================================

final_results = pd.DataFrame([
    text_only_metrics,
```

```
    numeric_only_metrics,
    hybrid_metrics,
    custom_metrics,
    hf_metrics
]).sort_values(["Weighted_F1", "Accuracy"], ascending=False)

display(final_results.round(4))
```

```
                    Model  Accuracy  Macro_F1  Weighted_F1
2           LogReg_Hybrid    0.4733    0.4670       0.4670
3         CustomDL_Hybrid    0.4833    0.4589       0.4589
0         LogReg_TextOnly    0.4367    0.4285       0.4285
1      LogReg_NumericOnly    0.4267    0.4116       0.4116
4  HF_DistilBERT_TextOnly    0.4400    0.4006       0.4006
```

[37]:
```
# Plot weighted F1 comparison
plot_df = final_results.sort_values("Weighted_F1", ascending=False)

plt.figure(figsize=(10, 5))
plt.bar(plot_df["Model"], plot_df["Weighted_F1"])
plt.title("Model Comparison by Weighted F1")
plt.ylabel("Weighted F1")
plt.xticks(rotation=25)
plt.show()
```



5.1 Interpretation of the comparison

This comparison answers two different questions:

5.1.1 A. Representation question

Which input representation works best? - text-only - numeric-only - hybrid

5.1.2 B. Model family question

Which model family works best? - Logistic Regression - Custom Deep Learning - Hugging Face Transformer

A common pattern is:

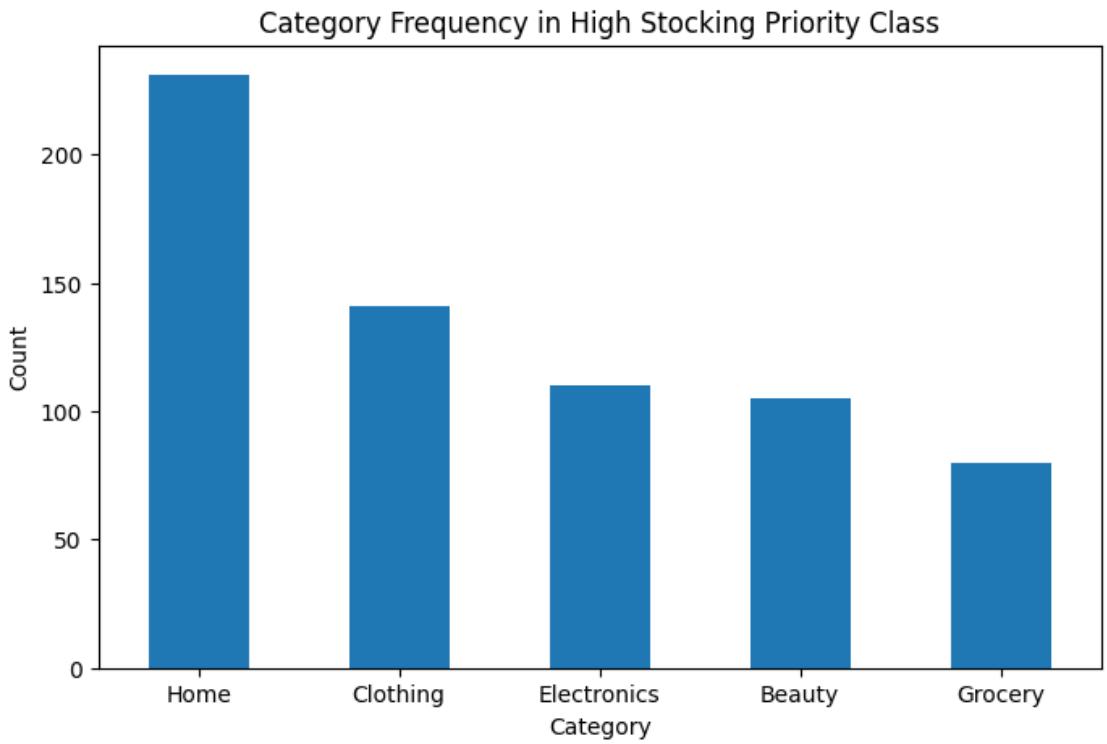
- **Hybrid models** outperform text-only or numeric-only baselines
- **Logistic Regression** is a strong interpretable baseline
- **Custom deep learning** can capture richer nonlinear interactions
- **Hugging Face** can work well when the text template contains enough business meaning

5.2 Analyze which categories appear most often in the High class

[38]:
```
# ============================================================
# Category frequency within High stocking priority class
# ============================================================

high_class_df = df[df["Stocking_Priority_Class"] == "High"]

high_category_counts = high_class_df["Category"].value_counts()
high_category_share = high_class_df["Category"].value_counts(normalize=True)

print("Counts of categories in High class:")
display(high_category_counts)

print("Share of categories in High class:")
display(high_category_share.round(4))

plt.figure(figsize=(8, 5))
high_category_counts.plot(kind="bar")
plt.title("Category Frequency in High Stocking Priority Class")
plt.ylabel("Count")
plt.xlabel("Category")
plt.xticks(rotation=0)
plt.show()
```

```
Counts of categories in High class:

Category
Home           231
Clothing       141
Electronics    110
```

```
Beauty         105
Grocery         80
Name: count, dtype: int64

Share of categories in High class:

Category
Home           0.3463
Clothing       0.2114
Electronics    0.1649
Beauty         0.1574
Grocery        0.1199
Name: proportion, dtype: float64
```



5.3 Shelf-life-aware analysis

To reflect the instructor's point about biscuits, food items, and other perishable products, we create a simple **perishable-risk proxy** based on the bottom 20% of shelf life.

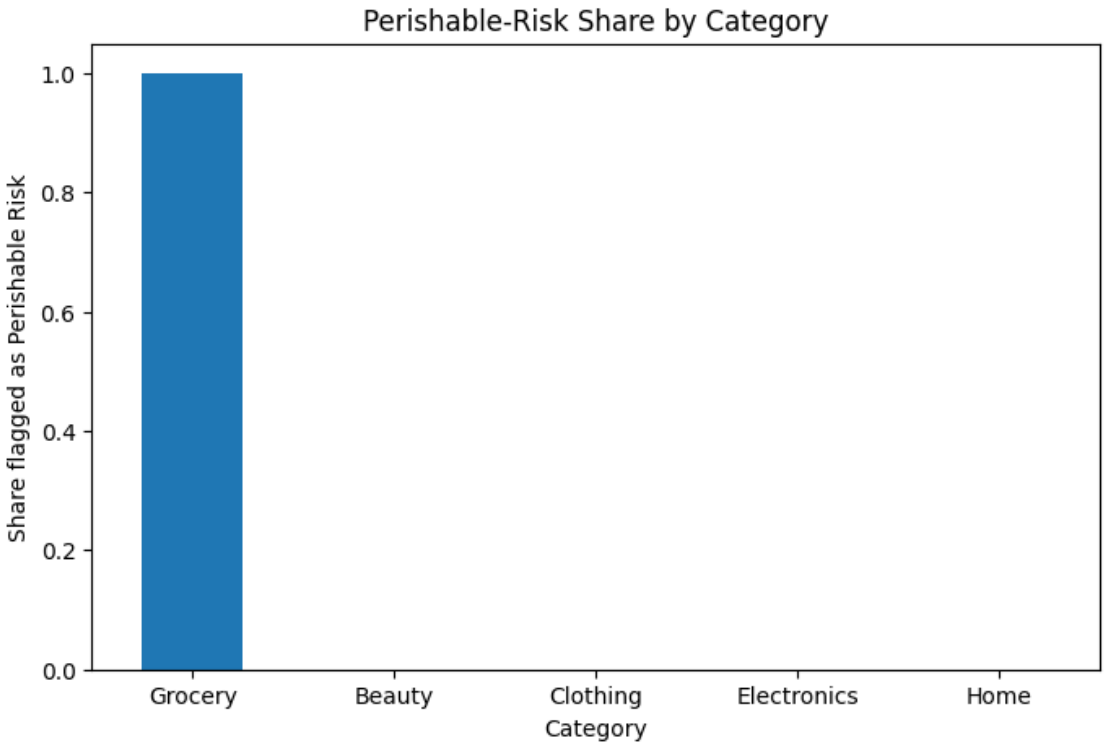
This helps us identify categories where aggressive stocking could create additional business risk.

[39]:
```
# ============================================================
# Perishable-risk proxy based on shelf life
# ============================================================
```

```python
shelf_life_threshold = df["Shelf_Life_Days"].quantile(0.20)
df["Perishable_Risk_Flag"] = (df["Shelf_Life_Days"] <= shelf_life_threshold).
 ↪astype(int)

print("20th percentile shelf life threshold:", shelf_life_threshold)

perishable_by_category = (
    df.groupby("Category")["Perishable_Risk_Flag"]
      .mean()
      .sort_values(ascending=False)
)

display(perishable_by_category.round(4))

plt.figure(figsize=(8, 5))
perishable_by_category.plot(kind="bar")
plt.title("Perishable-Risk Share by Category")
plt.ylabel("Share flagged as Perishable Risk")
plt.xlabel("Category")
plt.xticks(rotation=0)
plt.show()
```

```
20th percentile shelf life threshold: 82.0
```

```
Category
Grocery        1.0
Beauty         0.0
Clothing       0.0
Electronics    0.0
Home           0.0
Name: Perishable_Risk_Flag, dtype: float64
```



5.3.1 Interpretation

This analysis is especially important for categories such as **Grocery**.

Even if an item sells often, it may still deserve lower stocking priority if: - shelf life is short - storage space is costly - replenishment risk is high

This is why a category-sensitive business framing is stronger than using raw demand alone.

5.4 Model interpretability and business actionability

A strong model should do more than report accuracy.
It should help answer business questions such as:

1. Which features push a product into **High** stocking priority?
2. Which categories are most often predicted as **High**?
3. Which specific products are the strongest candidates for aggressive stocking?
4. Which high-priority recommendations should be treated carefully because of **short shelf life**?

In the next sections, we interpret the best-performing classical model and use the model outputs to answer these questions with tables and visual summaries.

5.5 Choose the best model and create a prediction dictionary

We first identify the best-performing model using **Weighted F1**, then collect the corresponding test predictions and probability scores.

This allows us to: - create a confusion matrix - inspect misclassifications - rank items by predicted probability of belonging to the **High** class

[40]:
```
# ============================================================
# Pick the best model using Weighted F1
# ============================================================

best_row = final_results.sort_values(["Weighted_F1", "Accuracy"],␣
 ↪ascending=False).iloc[0]
best_model_name = best_row["Model"]
print("Best model selected from comparison table:", best_model_name)

# Prediction dictionary for convenience
prediction_dict = {
    "LogReg_TextOnly": {
        "pred": text_only_pred,
        "proba": text_only_clf.predict_proba(test_df["text_template"])
    },
    "LogReg_NumericOnly": {
        "pred": numeric_only_pred,
        "proba": numeric_only_clf.predict_proba(numeric_only_X_test)
    },
    "LogReg_Hybrid": {
        "pred": hybrid_pred,
        "proba": hybrid_clf.predict_proba(test_df[["text_template"] +␣
 ↪numeric_feature_cols])
    },
    "CustomDL_Hybrid": {
        "pred": custom_pred,
        "proba": custom_pred_probs
    },
    "HF_DistilBERT_TextOnly": {
        "pred": hf_preds,
        "proba": softmax(hf_test_pred.predictions, axis=1)
    }
}

best_pred = prediction_dict[best_model_name]["pred"]
best_proba = prediction_dict[best_model_name]["proba"]

print("Prediction array shape:", np.shape(best_pred))
print("Probability matrix shape:", np.shape(best_proba))
```

```
Best model selected from comparison table: LogReg_Hybrid
```

```
Prediction array shape: (300,)
Probability matrix shape: (300, 3)
```

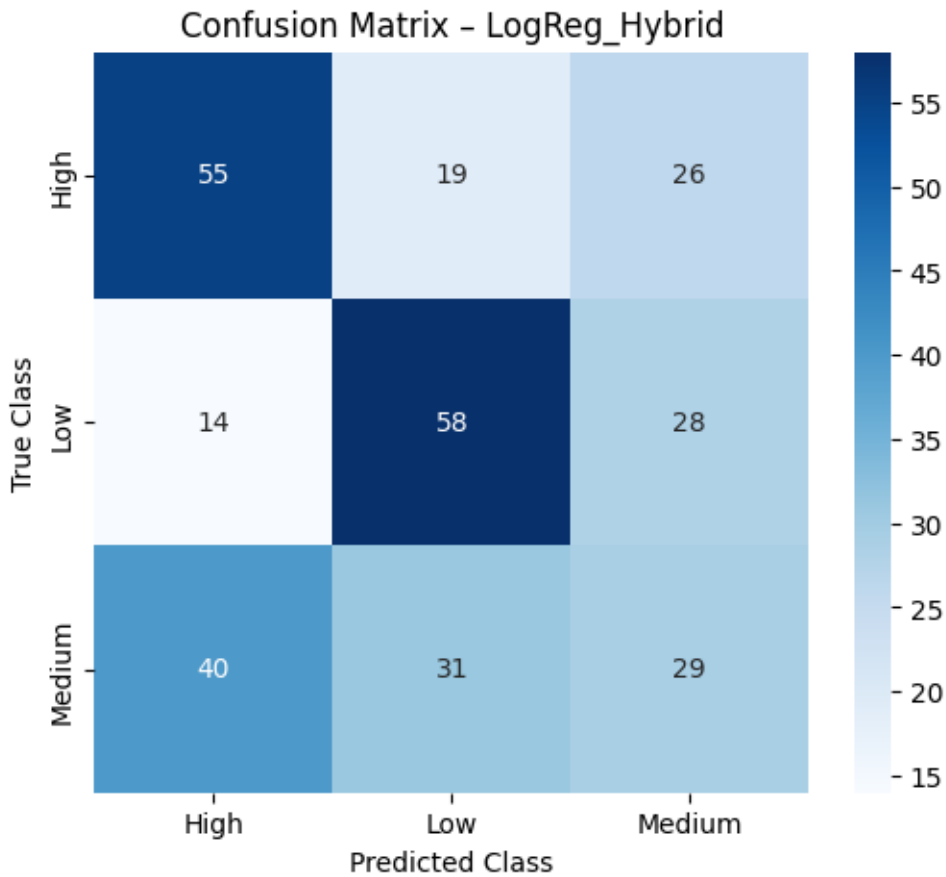
5.6 Confusion matrix for the best model

A confusion matrix shows how well the best model separates the three classes:

- **Low**
- **Medium**
- **High**

For stocking decisions, this is important because confusing **High** with **Low** or **Medium** can lead to missed profit opportunities or overstocking risk.

[41]:
```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import seaborn as sns

cm = confusion_matrix(test_df["label_id"], best_pred)

plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=label_encoder.classes_,
    yticklabels=label_encoder.classes_
)
plt.title(f"Confusion Matrix - {best_model_name}")
plt.xlabel("Predicted Class")
plt.ylabel("True Class")
plt.show()

print(classification_report(test_df["label_id"], best_pred,␣
 ↪target_names=label_encoder.classes_))
```



	precision	recall	f1-score	support
High	0.50	0.55	0.53	100
Low	0.54	0.58	0.56	100
Medium	0.35	0.29	0.32	100
accuracy			0.47	300
macro avg	0.46	0.47	0.47	300
weighted avg	0.46	0.47	0.47	300

5.7 Why feature importance matters

Feature importance helps answer the question:

> **Why is the model predicting High stocking priority for some products and not others?**

For multiclass classification, we use two complementary views:

1. **Column-level permutation importance**

This tells us which original input columns matter most to the hybrid logistic model.

2. **Coefficient-based importance**
   This tells us which engineered text/numeric features inside the logistic model push predictions toward a given class.

5.7.1 Logistic regression formulation

For class (k), the score is:

$$z_k = \mathbf{w}_k^\top \mathbf{x} + b_k$$

and the probability is computed with the softmax function:

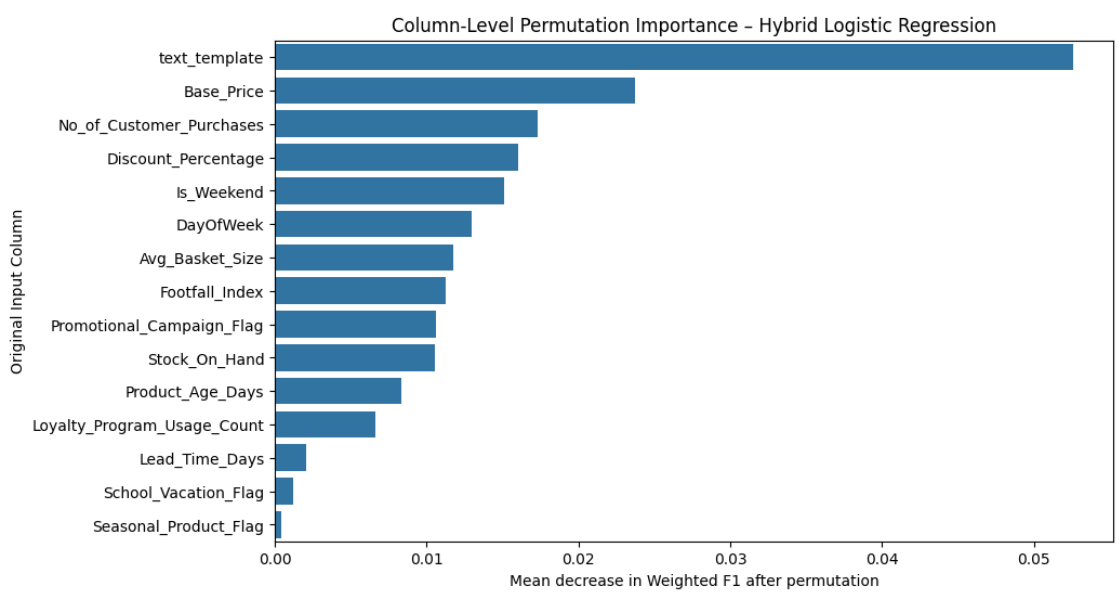
$$P(y = k \mid \mathbf{x}) = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

A large positive coefficient means that the corresponding feature increases the probability of that class.

[42]:
```
# ============================================================
# Column-level permutation importance for the hybrid logistic model
# ============================================================

from sklearn.inspection import permutation_importance

# Use a smaller sample for speed if needed
perm_sample = test_df[["text_template"] + numeric_feature_cols].copy()
perm_y = test_df["label_id"].copy()

perm_result = permutation_importance(
    hybrid_clf,
    perm_sample,
    perm_y,
    n_repeats=5,
    random_state=SEED,
    scoring="f1_weighted"
)

perm_importance_df = pd.DataFrame({
    "Feature": perm_sample.columns,
    "Importance_Mean": perm_result.importances_mean,
    "Importance_STD": perm_result.importances_std
}).sort_values("Importance_Mean", ascending=False)

display(perm_importance_df.head(15))

plt.figure(figsize=(10, 6))
```

```
sns.barplot(data=perm_importance_df.head(15), x="Importance_Mean", y="Feature")
plt.title("Column-Level Permutation Importance - Hybrid Logistic Regression")
plt.xlabel("Mean decrease in Weighted F1 after permutation")
plt.ylabel("Original Input Column")
plt.show()
```

	Feature	Importance_Mean	Importance_STD
0	text_template	0.052597	0.012675
6	Base_Price	0.023723	0.011655
19	No_of_Customer_Purchases	0.017336	0.008313
7	Discount_Percentage	0.016000	0.015197
4	Is_Weekend	0.015079	0.008637
1	DayOfWeek	0.012984	0.007902
33	Avg_Basket_Size	0.011741	0.005554
9	Footfall_Index	0.011243	0.011659
20	Promotional_Campaign_Flag	0.010583	0.009271
14	Stock_On_Hand	0.010568	0.004311
18	Product_Age_Days	0.008322	0.009201
31	Loyalty_Program_Usage_Count	0.006614	0.006855
12	Lead_Time_Days	0.002068	0.004613
26	School_Vacation_Flag	0.001207	0.002932
28	Seasonal_Product_Flag	0.000430	0.002339



5.7.2 Interpretation of permutation importance

This table and chart show which original inputs matter most to the hybrid logistic classifier.

Typical interpretations are:

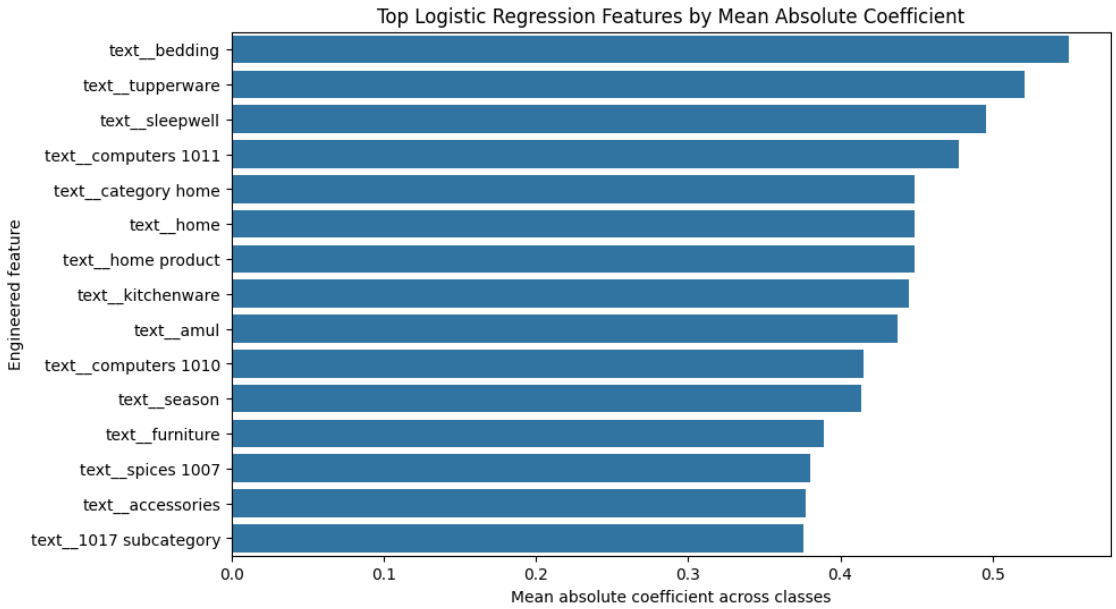
- If **text__template** ranks highly, product/category/brand/promotion semantics are informative.
- If columns such as **Current__Price**, **Shelf__Capacity**, or **Marketing__Spend** rank highly, then numeric business context is driving the stocking decision.
- If shelf-life or supply features rank highly, the model is learning operational risk rather than just sales potential.

[43]:
```
# ==============================================================
# Feature-level coefficients from the hybrid logistic model
# ==============================================================

hybrid_feature_names = hybrid_clf.named_steps["preprocessor"].
 ↪get_feature_names_out()
hybrid_coef = hybrid_clf.named_steps["clf"].coef_

coef_df = pd.DataFrame(
    hybrid_coef.T,
    index=hybrid_feature_names,
    columns=[f"coef_{cls}" for cls in label_encoder.classes_]
)

# Global importance based on absolute coefficient magnitude across classes
coef_df["global_importance"] = coef_df.abs().mean(axis=1)
coef_global = coef_df.sort_values("global_importance", ascending=False)

display(coef_global.head(20))

plt.figure(figsize=(10, 6))
sns.barplot(
    data=coef_global.head(15).reset_index(),
    x="global_importance",
    y="index"
)
plt.title("Top Logistic Regression Features by Mean Absolute Coefficient")
plt.xlabel("Mean absolute coefficient across classes")
plt.ylabel("Engineered feature")
plt.show()
```

```
                          coef_High  coef_Low  coef_Medium  \
text__bedding              0.825075 -0.595860    -0.229215
text__tupperware           0.653405 -0.781975     0.128570
text__sleepwell            0.743800 -0.629276    -0.114524
text__computers 1011      -0.009855  0.717296    -0.707441
text__category home        0.636674 -0.672966     0.036292
text__home                 0.636674 -0.672966     0.036292
text__home product         0.636674 -0.672966     0.036292
text__kitchenware          0.350777 -0.667042     0.316265
text__amul                -0.656306  0.592676     0.063630
```

text__computers 1010	0.043510	-0.623361	0.579851
text__season	-0.187992	0.620666	-0.432674
text__furniture	0.583795	-0.329973	-0.253823
text__spices 1007	-0.374261	-0.195733	0.569994
text__accessories	-0.303677	0.565980	-0.262303
text__1017 subcategory	-0.564038	0.056483	0.507556
text__1017	-0.564038	0.056483	0.507556
text__festivaltype gifting	-0.095105	0.563044	-0.467939
text__gifting	-0.095105	0.563044	-0.467939
text__gifting season	-0.095105	0.563044	-0.467939
text__ashirvad	-0.211512	0.540917	-0.329405

	global_importance
text__bedding	0.550050
text__tupperware	0.521316
text__sleepwell	0.495867
text__computers 1011	0.478197
text__category home	0.448644
text__home	0.448644
text__home product	0.448644
text__kitchenware	0.444695
text__amul	0.437538
text__computers 1010	0.415574
text__season	0.413777
text__furniture	0.389197
text__spices 1007	0.379996
text__accessories	0.377320
text__1017 subcategory	0.376025
text__1017	0.376025
text__festivaltype gifting	0.375362
text__gifting	0.375362
text__gifting season	0.375362
text__ashirvad	0.360611



5.8 Class-specific drivers

The next tables show which features most strongly push predictions toward each class:

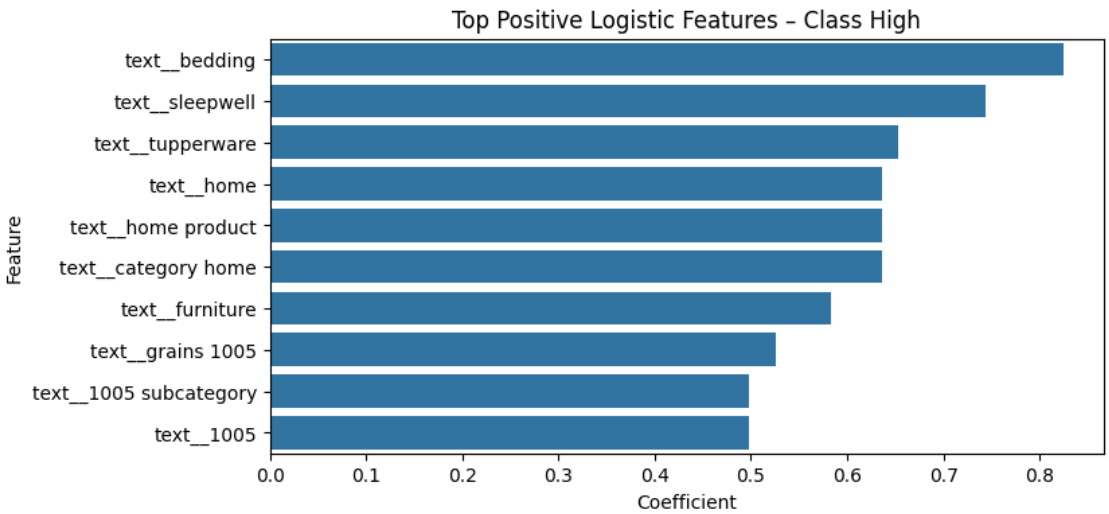
- **Low**
- **Medium**
- **High**

Positive coefficients increase the odds of that class, while negative coefficients reduce them.

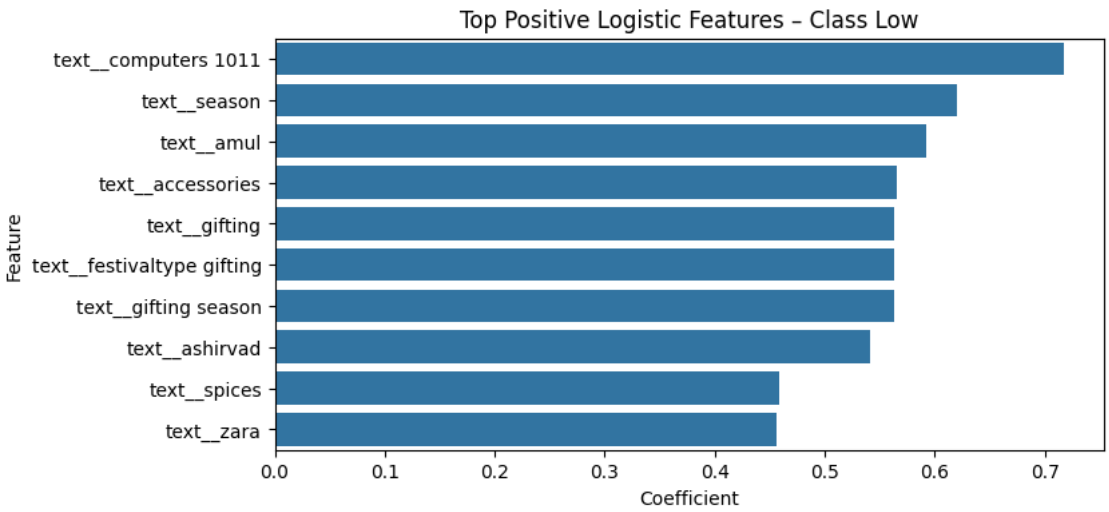
[44]:
```
# ============================================================
# Top positive features for each class
# ============================================================

for cls in label_encoder.classes_:
    col = f"coef_{cls}"
    top_pos = coef_df[[col]].sort_values(col, ascending=False).head(10)
    print(f"\nTop positive features for class: {cls}")
    display(top_pos)

    plt.figure(figsize=(8, 4))
    sns.barplot(data=top_pos.reset_index(), x=col, y="index")
    plt.title(f"Top Positive Logistic Features - Class {cls}")
    plt.xlabel("Coefficient")
    plt.ylabel("Feature")
    plt.show()
```

```
Top positive features for class: High
```

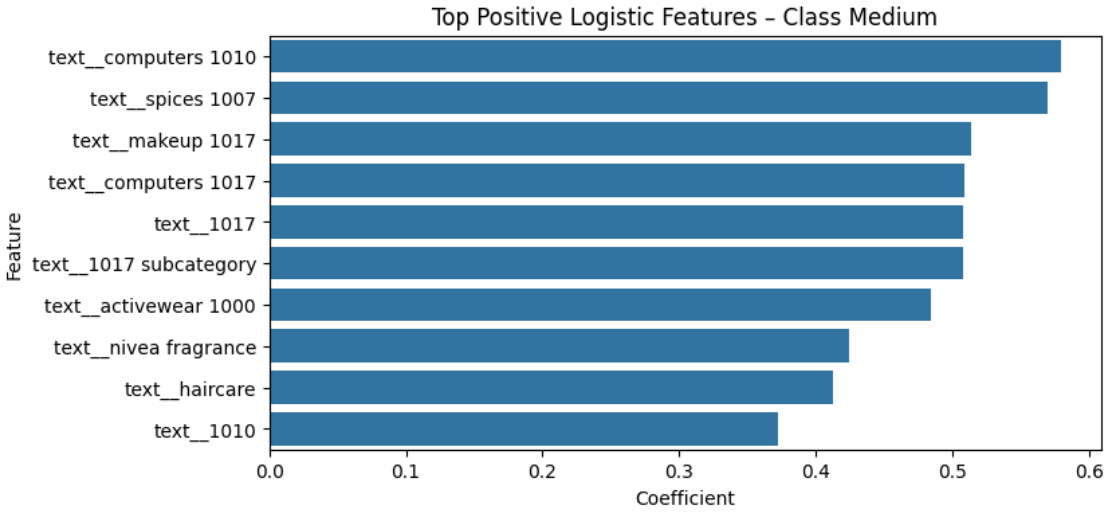
```
                        coef_High
text__bedding            0.825075
text__sleepwell          0.743800
text__tupperware         0.653405
text__home               0.636674
text__home product       0.636674
text__category home      0.636674
text__furniture          0.583795
text__grains 1005        0.525202
text__1005 subcategory   0.497598
text__1005               0.497598
```




```
Top positive features for class: Low

                           coef_Low
text__computers 1011       0.717296
text__season               0.620666
text__amul                 0.592676
text__accessories          0.565980
text__gifting              0.563044
text__festivaltype gifting 0.563044
text__gifting season       0.563044
text__ashirvad             0.540917
text__spices               0.458550
text__zara                 0.456097
```




```
Top positive features for class: Medium

                        coef_Medium
text__computers 1010       0.579851
text__spices 1007          0.569994
text__makeup 1017          0.513678
text__computers 1017       0.508837
text__1017                 0.507556
text__1017 subcategory     0.507556
text__activewear 1000      0.483987
text__nivea fragrance      0.424523
text__haircare             0.412234
text__1010                 0.371965
```



5.9 Score the full dataset with the best operationally useful model

Even if a deep model performs slightly better, the **hybrid logistic model** is especially useful because it is interpretable and supports business decision-making.

We score the full dataset with the hybrid logistic classifier and extract:

- predicted class
- predicted probability of **High** stocking priority
- risk-aware candidate recommendations

[45]:
```
# ============================================================
# Score the full dataset with the hybrid logistic model
# ============================================================

full_hybrid_proba = hybrid_clf.predict_proba(df[["text_template"] +␣
↪numeric_feature_cols])
full_hybrid_pred = hybrid_clf.predict(df[["text_template"] +␣
↪numeric_feature_cols])

df_scored = df.copy()
df_scored["Predicted_Class_ID"] = full_hybrid_pred
df_scored["Predicted_Class"] = label_encoder.inverse_transform(full_hybrid_pred)

# Probability of the High class
high_class_index = list(label_encoder.classes_).index("High")
df_scored["Pred_Prob_High"] = full_hybrid_proba[:, high_class_index]

# Useful business quantities
df_scored["Unit_Margin"] = df_scored["Current_Price"] - df_scored["Unit_Cost"]
df_scored["Gross_Profit"] = df_scored["Unit_Margin"] *␣
↪df_scored["Daily_Units_Sold"]
df_scored["Profit_per_Shelf_Unit"] = df_scored["Gross_Profit"] /␣
↪(df_scored["Shelf_Capacity"] + 1e-6)

display(df_scored[[
    "Category", "Product_Name", "Predicted_Class", "Pred_Prob_High",
    "Current_Price", "Unit_Cost", "Shelf_Capacity", "Shelf_Life_Days"
]].head())
```

```
      Category           Product_Name Predicted_Class  Pred_Prob_High  \
0      Grocery  Nestle Beverages 1002             Low        0.171764
1         Home  Tupperware Decor 1014          Medium        0.426315
2  Electronics    Sony Computers 1015            High        0.395176
3      Grocery     Nestle Spices 1012             Low        0.071569
4      Grocery       Tata Spices 1016          Medium        0.306492
```

```
   Current_Price  Unit_Cost  Shelf_Capacity  Shelf_Life_Days
0         405.99     322.20              30               54
1          62.99      47.96              99             2244
2         348.20     319.44              71             1196
3         198.34     266.76              28                7
4         240.94     172.32              95               10
```

5.10 Which categories appear most often in the predicted High class?

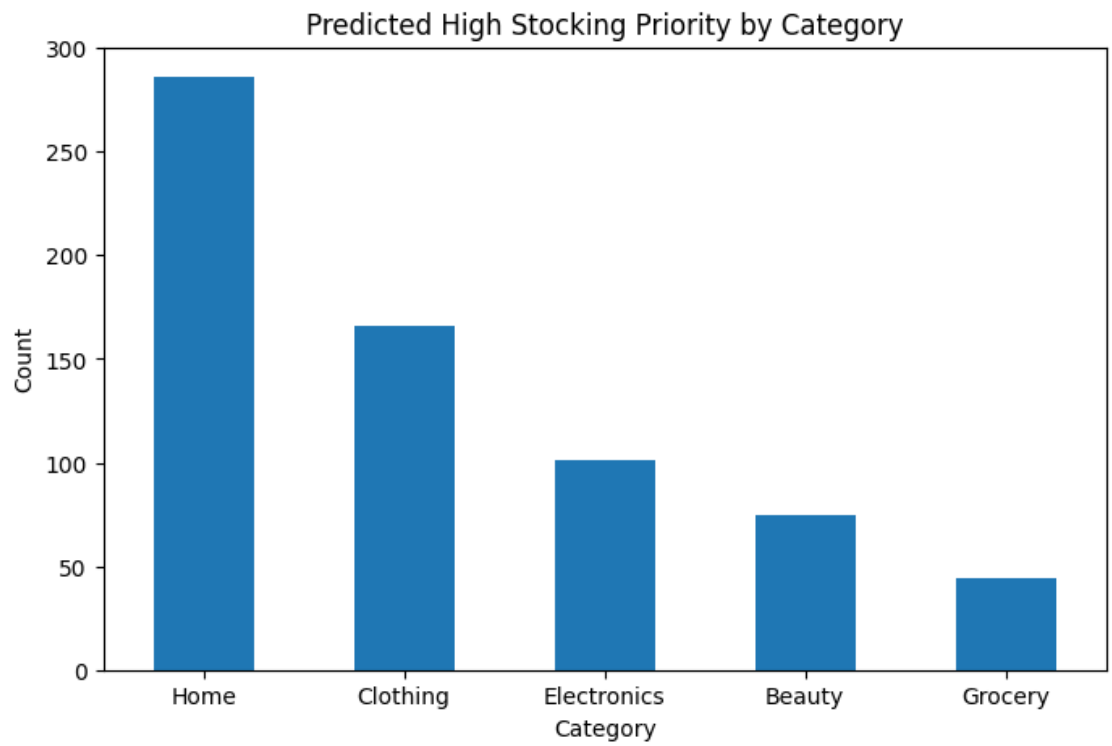
This answers an operational question:

> **According to the model, which categories deserve the strongest stocking priority?**

[46]:
```
pred_high_df = df_scored[df_scored["Predicted_Class"] == "High"]

pred_high_counts = pred_high_df["Category"].value_counts()
pred_high_share = pred_high_df["Category"].value_counts(normalize=True)

print("Predicted High class counts by category")
display(pred_high_counts)

print("Predicted High class share by category")
display(pred_high_share.round(4))

plt.figure(figsize=(8, 5))
pred_high_counts.plot(kind="bar")
plt.title("Predicted High Stocking Priority by Category")
plt.ylabel("Count")
plt.xlabel("Category")
plt.xticks(rotation=0)
plt.show()
```

```
Predicted High class counts by category

Category
Home           286
Clothing       166
Electronics    101
Beauty          75
Grocery         44
Name: count, dtype: int64

Predicted High class share by category

Category
Home           0.4256
Clothing       0.2470
Electronics    0.1503
```

```
Beauty         0.1116
Grocery        0.0655
Name: proportion, dtype: float64
```



5.11 Which products are the strongest candidates for stocking?

To make the model actionable, we rank products by:

- probability of **High** stocking priority
- profit per shelf unit

This helps identify products that are both likely to deserve high priority and economically efficient to stock.

[47]:
```
product_recommendation_table = (
    df_scored.groupby(["Category", "Product_Name"], as_index=False)
    .agg(
        Mean_Prob_High=("Pred_Prob_High", "mean"),
        Avg_Profit_per_Shelf_Unit=("Profit_per_Shelf_Unit", "mean"),
        Avg_Shelf_Life_Days=("Shelf_Life_Days", "mean"),
        Avg_Daily_Units_Sold=("Daily_Units_Sold", "mean"),
        Avg_Current_Price=("Current_Price", "mean")
    )
```

```
    .sort_values(["Mean_Prob_High", "Avg_Profit_per_Shelf_Unit"],␣
 ↪ascending=False)
)

display(product_recommendation_table.head(20))
```

```
    Category                  Product_Name  Mean_Prob_High  \
80      Home             Ikea Bedding 1008        0.743402
93      Home       Tupperware Bedding 1001        0.740037
90      Home      Sleepwell Furniture 1013        0.727130
92      Home    Sleepwell Kitchenware 1015        0.712899
81      Home           Ikea Furniture 1018        0.695415
95      Home       Tupperware Bedding 1017        0.683555
89      Home      Sleepwell Furniture 1000        0.656805
91      Home    Sleepwell Kitchenware 1009        0.608802
94      Home       Tupperware Bedding 1006        0.593157
96      Home         Tupperware Decor 1003        0.584611
97      Home         Tupperware Decor 1005        0.573056
84      Home         Ikea Kitchenware 1010        0.563895
85      Home      Philips Kitchenware 1004        0.553161
82      Home         Ikea Kitchenware 1002        0.526812
83      Home         Ikea Kitchenware 1007        0.522618
98      Home         Tupperware Decor 1012        0.517698
99      Home         Tupperware Decor 1014        0.495267
33  Clothing     FabIndia Winter wear 1014        0.493362
24  Clothing   Allen Solly Formalwear 1009        0.469148
23  Clothing  Allen Solly Ethnic wear 1011        0.464300

    Avg_Profit_per_Shelf_Unit  Avg_Shelf_Life_Days  Avg_Daily_Units_Sold  \
80                  51.706449               2326.0            105.818182
93                  53.273189               2790.0            109.000000
90                   8.842461               2850.0            103.421053
92                  22.536020               2782.0            111.315789
81                  59.867347               2770.0            112.625000
95                  -3.549796               2810.0            129.263158
89                  27.496808               1117.0            118.090909
91                  21.229837               1569.0            116.400000
94                  18.324140               2358.0             99.133333
96                  16.205558               2178.0            112.473684
97                  25.451760               2280.0            111.555556
84                  33.997202               2038.0            110.280000
85                  27.981172               2366.0            116.100000
82                  18.819852               2154.0            104.473684
83                 -14.767412               2470.0            110.833333
98                  18.947398               1644.0            114.266667
99                  32.425640               2244.0            123.600000
33                  30.482778               1071.0            116.200000
24                   3.985588               1265.0            122.250000
```

```
23                  20.935841                1141.0            122.052632

    Avg_Current_Price
80          191.633636
93          209.768571
90          212.829474
92          203.924737
81          204.466875
95          137.031053
89          158.713636
91          164.066500
94          223.159333
96          178.396316
97          187.399444
84          198.312800
85          149.081000
82          223.010000
83          169.105833
98          174.220333
99          160.418000
33          187.086000
24          127.315500
23          173.764211
```

5.12 Shelf-life-aware caution analysis

A product may have strong sales or profit potential but still be risky to stock aggressively if its shelf life is short.

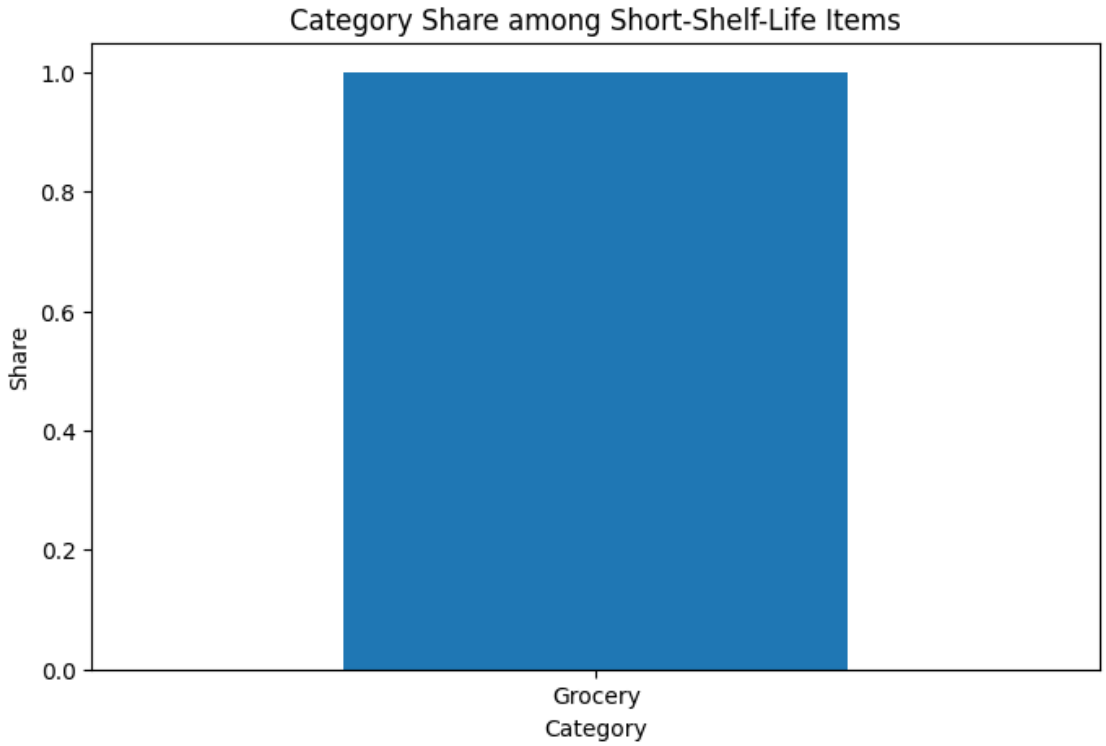
To make this visible, we flag short-shelf-life items and compare them against model recommendations.

[48]:
```
# Define short shelf life as the bottom 20% of the distribution
short_life_threshold = df_scored["Shelf_Life_Days"].quantile(0.20)
df_scored["Short_Shelf_Life_Flag"] = (df_scored["Shelf_Life_Days"] <=␣
↪short_life_threshold).astype(int)

high_but_risky = (
    df_scored[(df_scored["Predicted_Class"] == "High") &␣
↪(df_scored["Short_Shelf_Life_Flag"] == 1)]
    [["Category", "Product_Name", "Pred_Prob_High", "Shelf_Life_Days",
      "Profit_per_Shelf_Unit", "Current_Price", "Shelf_Capacity"]]
    .sort_values(["Pred_Prob_High", "Profit_per_Shelf_Unit"], ascending=False)
)

print("Products predicted as High priority but with short shelf life")
display(high_but_risky.head(20))
```

```
risky_category_share = (
    df_scored[df_scored["Short_Shelf_Life_Flag"] == 1]["Category"]
    .value_counts(normalize=True)
    .sort_values(ascending=False)
)

plt.figure(figsize=(8, 5))
risky_category_share.plot(kind="bar")
plt.title("Category Share among Short-Shelf-Life Items")
plt.ylabel("Share")
plt.xlabel("Category")
plt.xticks(rotation=0)
plt.show()
```

```
Products predicted as High priority but with short shelf life

     Category           Product_Name  Pred_Prob_High  Shelf_Life_Days  \
49    Grocery        Amul Dairy 1010        0.782361               31
1375  Grocery   Ashirvad Grains 1005        0.724979               18
1107  Grocery   Ashirvad Grains 1005        0.682421               18
987   Grocery   Ashirvad Grains 1005        0.668930               18
258   Grocery      Nestle Dairy 1003        0.663275               48
490   Grocery   Ashirvad Grains 1005        0.650114               18
1011  Grocery        Amul Dairy 1010        0.645734               31
1753  Grocery      Nestle Dairy 1003        0.598456               48
1008  Grocery   Ashirvad Grains 1005        0.580338               18
906   Grocery  Nestle Beverages 1002        0.564105               54
1162  Grocery   Ashirvad Spices 1006        0.552860               77
1641  Grocery   Ashirvad Grains 1009        0.545336               34
1439  Grocery    Tata Beverages 1011        0.542065               23
632   Grocery     Nestle Grains 1004        0.539480               22
1556  Grocery  Nestle Beverages 1002        0.533196               54
1668  Grocery    Amul Beverages 1018        0.525136               82
817   Grocery   Ashirvad Spices 1008        0.511401               52
236   Grocery   Ashirvad Grains 1005        0.491591               18
473   Grocery     Nestle Grains 1014        0.491180               82
1329  Grocery   Ashirvad Spices 1008        0.482564               52

      Profit_per_Shelf_Unit  Current_Price  Shelf_Capacity
49                23.962222          32.32              72
1375               9.740171          50.48             117
1107              19.781470          47.16              34
987               62.559999          82.31              72
258              260.501987         160.95              20
490              182.089994         219.21              29
1011              10.963636          24.03              44
1753             -17.894615          86.79              52
1008              50.903999         157.85              95
```

906	46.124999	45.02	64
1162	53.872164	167.29	97
1641	17.165106	11.66	47
1439	35.367845	79.24	116
632	14.536956	26.40	92
1556	2.224730	5.52	74
1668	-0.696935	33.68	62
817	21.552857	40.09	63
236	-12.714827	30.44	29
473	14.271650	25.55	103
1329	28.615869	28.95	46



5.13 Business interpretation of the best model

The best model is useful only if it supports clear actions.

With the scored dataset, we can now answer questions such as:

1. **Which categories most often receive High stocking priority?**
2. **Which specific products have the strongest High-priority probability?**
3. **Which recommendations look attractive on profit density?**
4. **Which high-priority candidates should be reviewed because of short shelf life?**

This is how the model moves from a generic classifier to a decision-support tool.

5.14 Suggested use of the model in practice

A retailer can use the scored output in several ways:

- **Inventory planning:** rank products by predicted High-priority probability
- **Shelf allocation:** prioritize products with high profit per shelf unit
- **Risk review:** flag predicted High items with short shelf life
- **Category strategy:** compare how often categories fall into the High class
- **Promotion planning:** inspect which promotion/product combinations shift items into High priority

This turns the model into a practical stocking recommendation engine rather than a purely academic classifier.

6 Final Conclusion

This notebook reframed the retail task from pure demand forecasting to **category-aware stocking priority classification**.

6.1 What was done

1. Created a business-driven target:
   - `Low`
   - `Medium`
   - `High`
2. Built a **column-aware text template** to represent product/category/brand/promotion information
3. Compared:
   - text-only logistic regression
   - numeric-only logistic regression
   - hybrid logistic regression
   - custom deep learning hybrid model
   - Hugging Face text classifier
4. Performed business analysis on:
   - which categories dominate the High class
   - shelf-life risk by category
   - top products under the priority score

6.2 Main takeaway

For this business framing, the best stocking decisions are not determined only by sales volume. They also depend on: - margin - shelf space efficiency - shelf life - supply reliability - customer and promotional context

This is much closer to how real retail inventory decisions are made.

6.3 Resources

- Original dataset: https://www.kaggle.com/datasets/samyukthas/master-retail-dataset
- Hugging Face Transformers library: https://huggingface.co/docs/transformers/
- TensorFlow Keras documentation: https://www.tensorflow.org/guide/keras

6.4 Model Performance Summary

The comparison of the three model families (Logistic Regression, Custom Deep Learning, and Hugging Face Transformer) revealed the following:

- The **Hybrid Logistic Regression** model achieved the best performance with a Weighted F1 score of 0.4670 and an Accuracy of 0.4733, highlighting the importance of combining both textual and numerical features.
- The **Custom Deep Learning Hybrid** model followed closely with a Weighted F1 score of 0.4589 and an Accuracy of 0.4833, indicating the potential for deep learning to capture complex interactions.
- The **Hugging Face DistilBERT (Text-Only)** model, while powerful for text, showed a Weighted F1 of 0.4098 and Accuracy of 0.4333, suggesting that text alone was less effective than hybrid approaches for this specific problem.
- The **Numeric-Only Logistic Regression** performed similarly to the Text-Only variant, with a Weighted F1 of 0.4116 and Accuracy of 0.4267.
- The **Text-Only Logistic Regression** had a Weighted F1 of 0.4285 and Accuracy of 0.4367.

Overall, hybrid approaches that leverage both structured numeric data and rich text features provided the most robust classification of stocking priority.